Sources of magnetic field

Ampere's law

→ BIOT-SAVART LAW

→ long wire and interaction between 2 wires

→ magnetic field of a single loop, magnetic dipole

→ electric dipole vs magnetic dipole, toque, PE

→ electric field vs magnetic field

→ Ampere's law , Ampere

→ Solenoid and torroid

→ Maxwell equations for constant magnetic and electric field.

Oersted experiment:

Monopoles don't exist → What is the source of magnetism

(charges are the source of electric field)

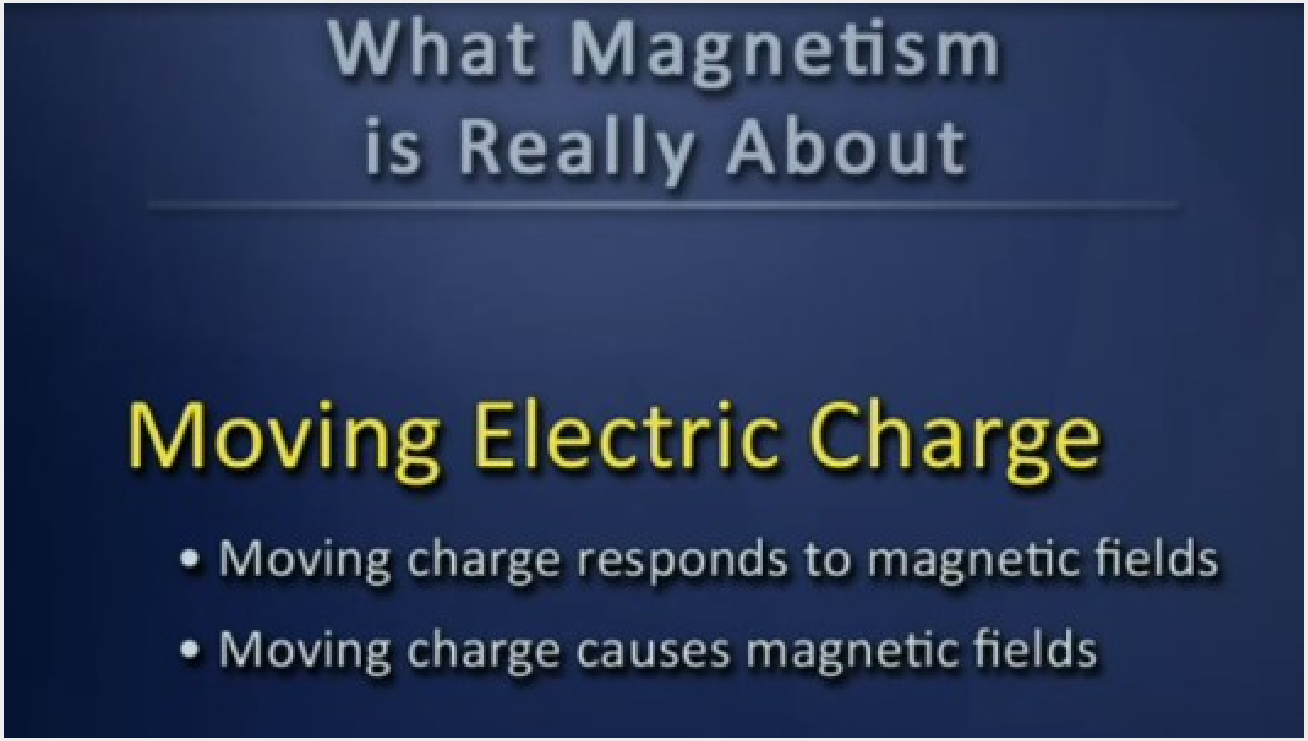
Magnetic Monopoles?

Magnetic monopoles: isolated south and north magnetic poles

- Magnetic analogs of electric charge
- Have never been found
- Theories of early universe suggest they might exist
- In the absence of magnetic monopoles:
  - The net number of magnetic field lines emerging from a closed surface is zero

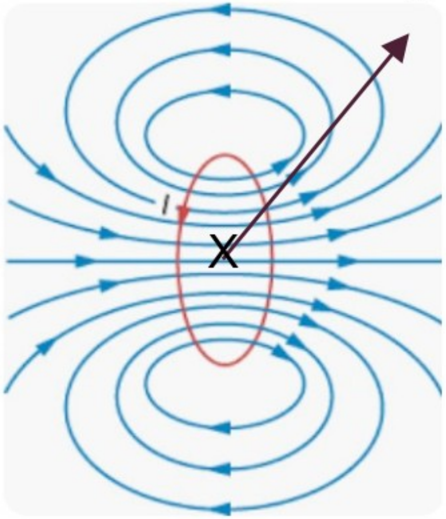
Gauss's Law for Magnetism $\oint \vec{B} \bullet d\vec{A} = 0$

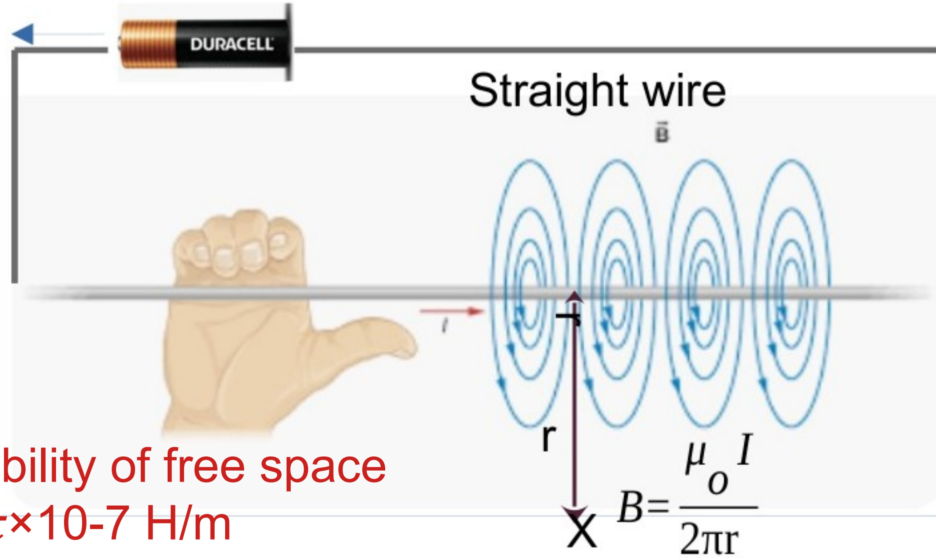
The source of magnetism is moving charges



loop/few loops

$$B=\frac{\mu_o I}{2R}$$

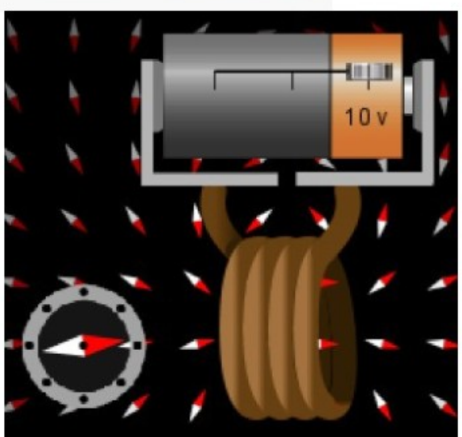




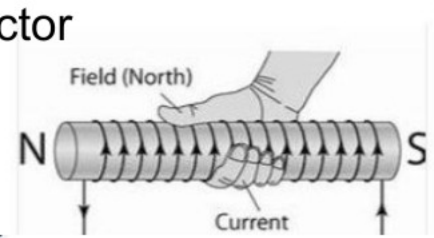
$$B=\frac{\mu_o I}{2\pi r}$$

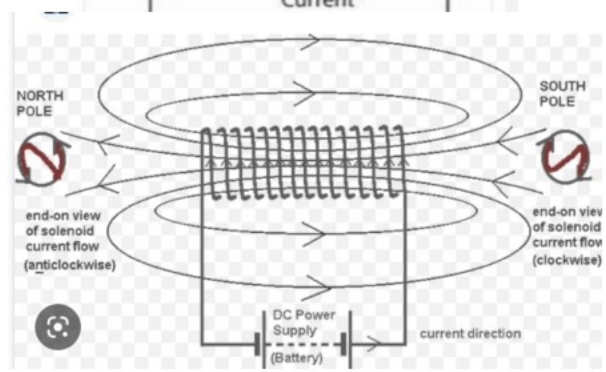
permeability of free space
$\mu_0 = 4\pi\times10^{-7}$ H/m
1.3×10^{-6} m kg s^{-2} A^{-2}

Where is the North?



solenoid/coil/inductor





$$B=\mu_o nI$$

B Inside is uniform $n=N/L$
Outside is almost 0 $r<<L$

In the equations describing electric and magnetic fields and their propagation, three constants are normally used. One is the speed of light c, and the other two are the electric permittivity of free space ε_0 and the magnetic permeability of free space, μ_0. The magnetic permeability of free space is taken to have the exact value

$$\mu_0 = 4\pi \times 10^{-7} N / A^2$$

This contains the force unit N for Newton and the unit A is the Ampere, the unit of electric current.

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$$\varepsilon_0 = 8.854 \cdot 10^{-12} \quad \text{[Farads/meter]}$$

$$\varepsilon_0 = \frac{1}{\mu_0 c^2}$$

ε_0 = vacuum permittivity

μ_0 = magnetic constant

c = speed of light

$$k_e = \frac{1}{4\pi\epsilon_0}$$

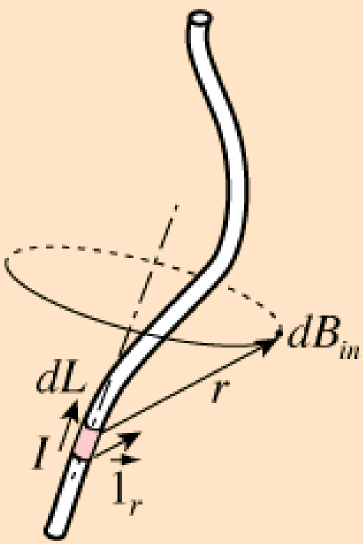
$Ke = 9 \times 10^9$

How to find the magentic field at a given point. Given A current distribution → use calculus and Biot-Savart law.

Biot-Savart Law

From hyperphysics

The Biot-Savart Law relates magnetic fields to the currents which are their sources. In a similar manner, Coulomb's law relates electric fields to the point charges which are their sources. Finding the magnetic field resulting from a current distribution involves the vector product, and is inherently a calculus problem when the distance from the current to the field point is continuously changing.



Magnetic field of a current element

$$d\vec{B} = \frac{\mu_0 I d\vec{L} \times \vec{1}_r}{4\pi r^2}$$

where

$d\vec{L}$ = infinitesmal length of conductor carrying electric current I

$\vec{1}_r$ = unit vector to specify the direction of the the vector distance r from the current to the field point.

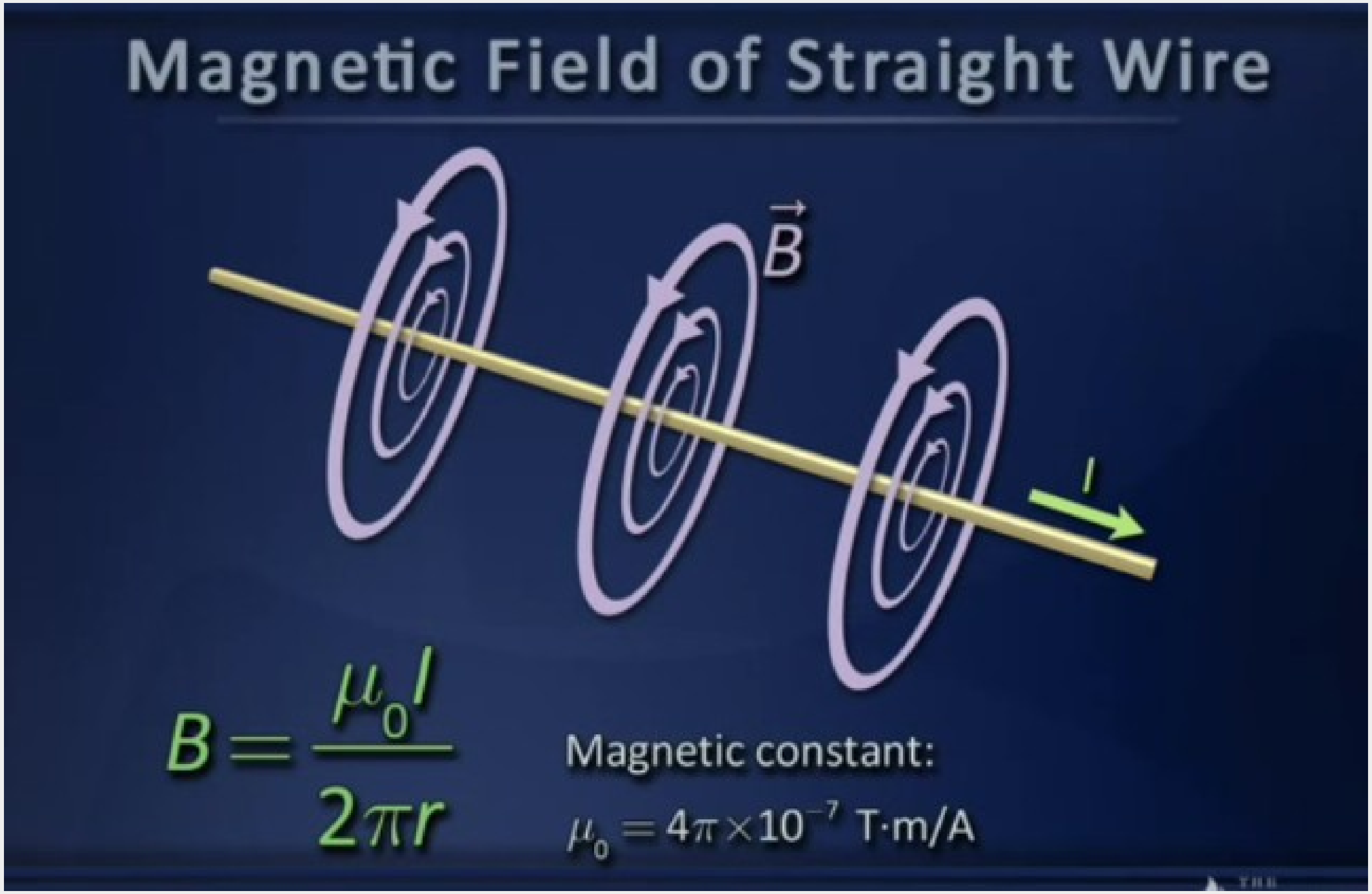
See the magnetic field sketched for the straight wire to see the geometry of the magnetic field of a current.

The Biot-Savart law describes the magnetic field (B) generated by a small current element (Idl) at a point P due to that element.

B falls as 1/r^2 but only for a small Segment dL.

Then you need to integrate over the Whole circuit.

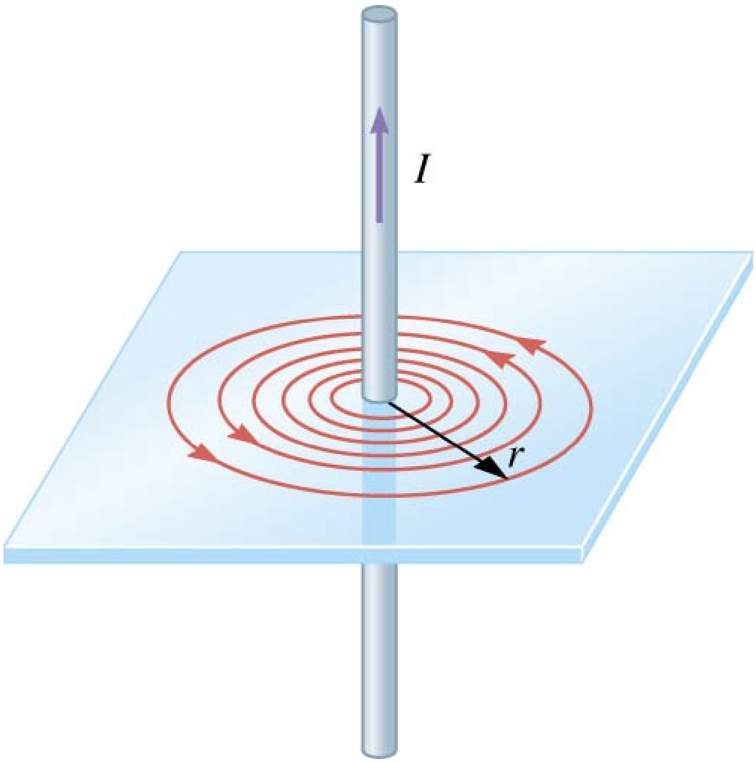
Using Biot-Savart law



The B falls as 1/r.

https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_University_Physics_(OpenStax)/Book%3A_University_Physics_II_-_Thermodynamics_Electricity_and_Magnetism_(OpenStax)/12%3A_Sources_of_Magnetic_Fields/12.03%3A_Magnetic_Field_due_to_a_Thin_Straight_Wire

APPLICATION of BIOT-SAVART : A LONG, STRAIGHT WIRE

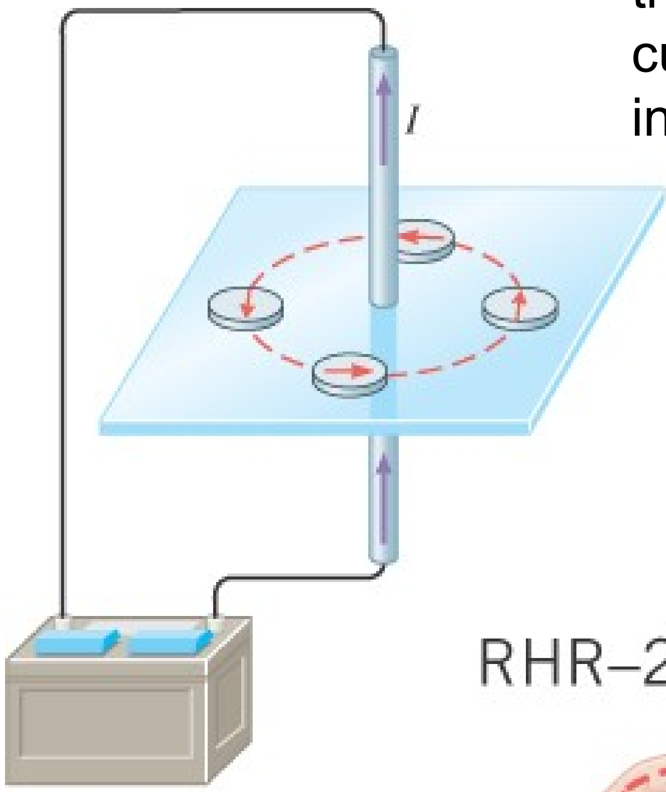


$$B = \frac{\mu_o I}{2\pi r}$$

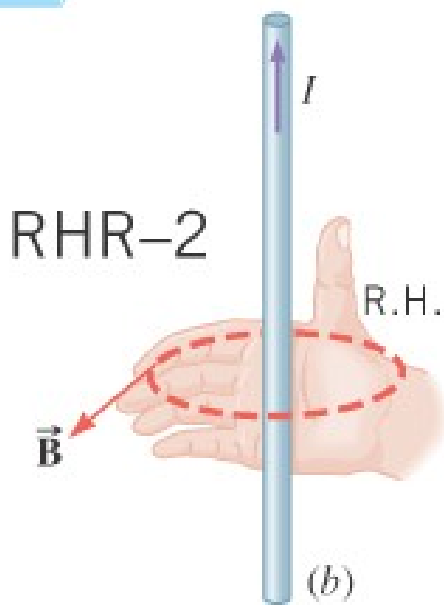
$$\mu_o = 4\pi \times 10^{-7} \, T \cdot m / A$$

permeability of
free space

Right-Hand Rule No. 2. Curl the fingers of the right hand into the shape of a half-circle. Point the thumb in the direction of the conventional current, and the tips of the fingers will point in the direction of the magnetic field.



(a)



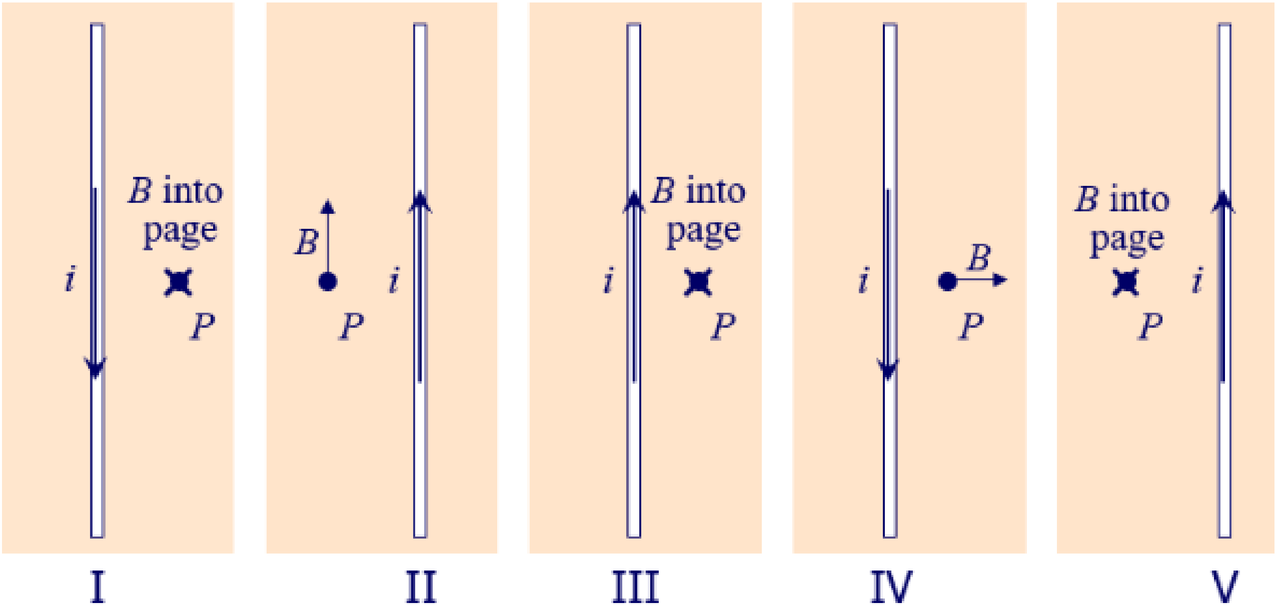
(b)

C

Direction of Magnetic Field

1. Which drawing below shows the correct direction of the magnetic field, B, at the point P?

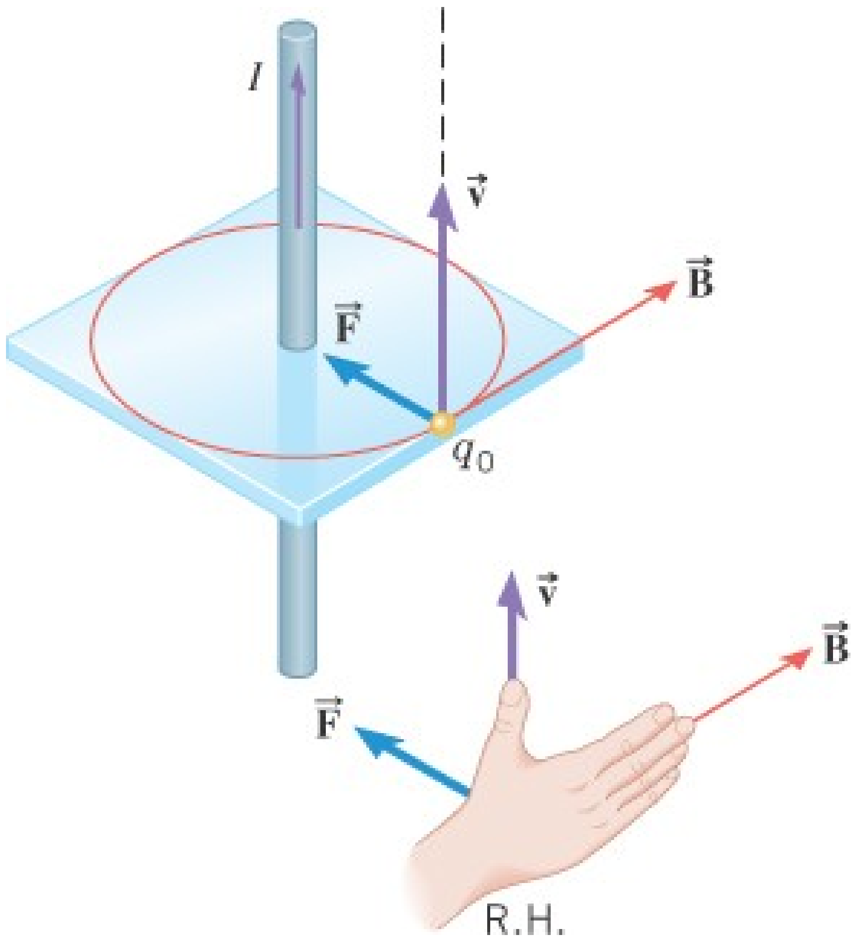
A. I.
B. II.
C. III.
D. IV.



Example 7 A Current Exerts a Magnetic Force on a Moving Charge

The long straight wire carries a current of 3.0 A. A particle has a charge of $+6.5\text{x}10^{-6}$ C and is moving parallel to the wire at a distance of 0.050 m. The speed of the particle is 280 m/s.

Determine the magnitude and direction of the magnetic force on the particle.

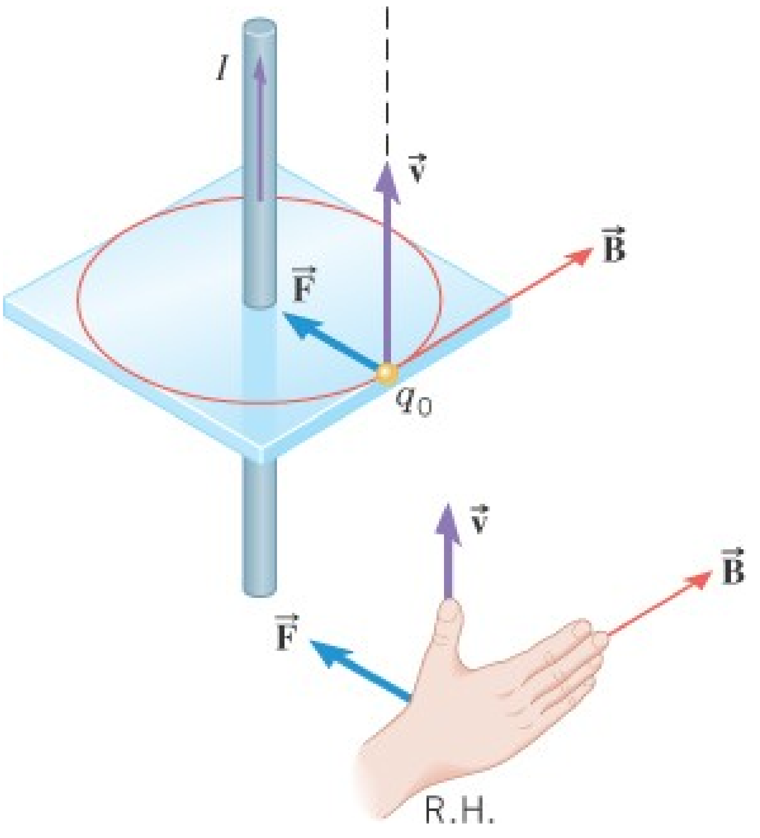


Step1: find B

Step2: find magnetic force

$$F = qvB\sin\theta = qv\left(\frac{\mu_o I}{2\pi r}\right)\sin\theta$$

$$B = \frac{\mu_o I}{2\pi r}$$



Three wires sit at the corners of a square, all carrying currents of 2 amps into the page as shown in Figure 12.3.4. Calculate the magnitude of the magnetic field at the other corner of the square, point **P**, if the length of each side of the square is 1 cm.

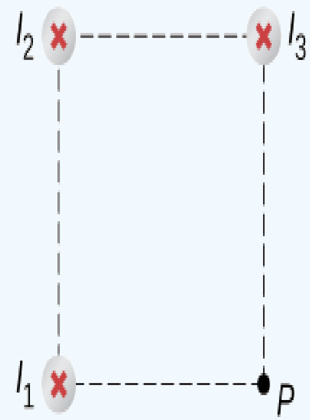
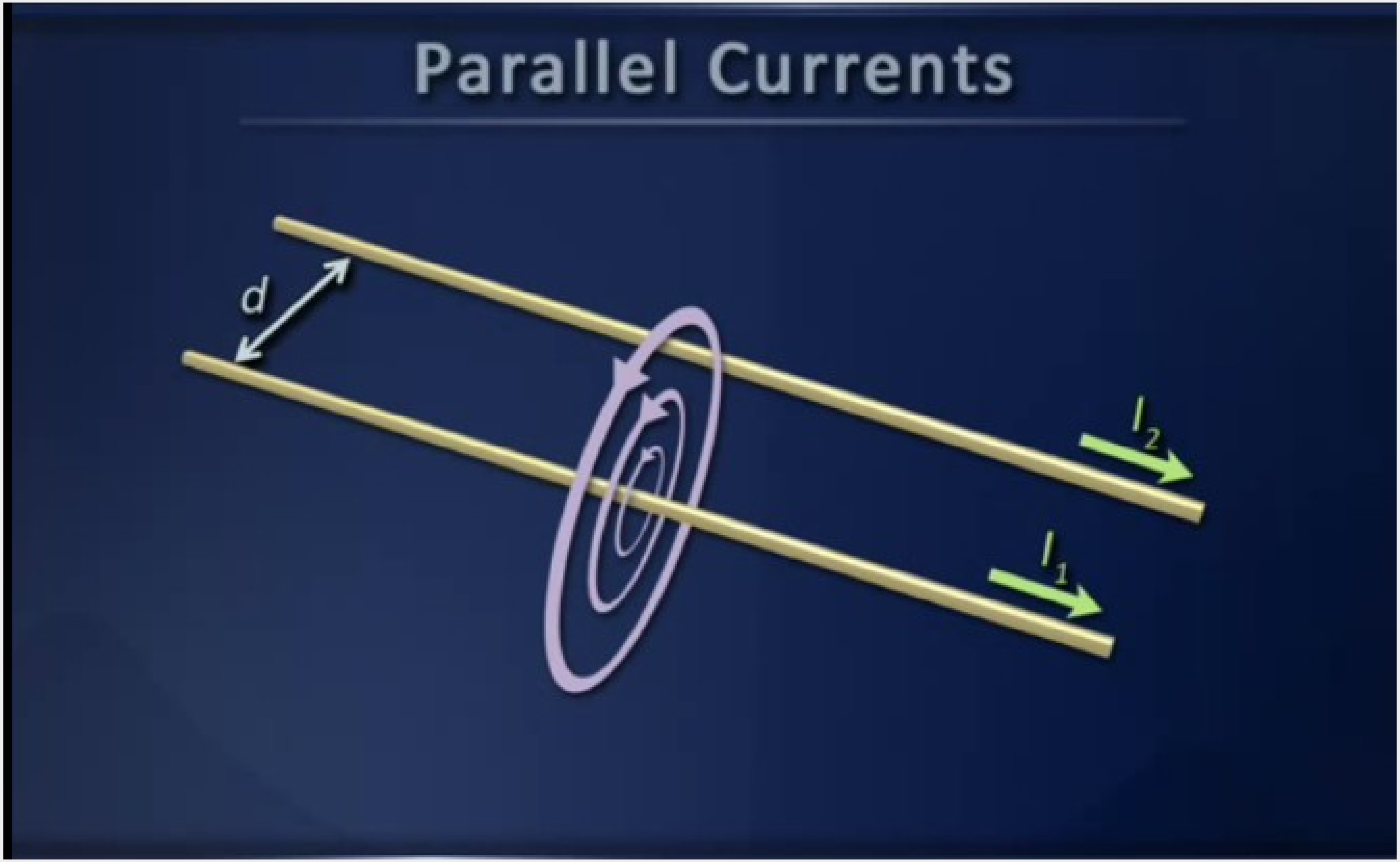


Figure 12.3.4: Three wires have current flowing into the page. The magnetic field is determined at the fourth corner of the square.

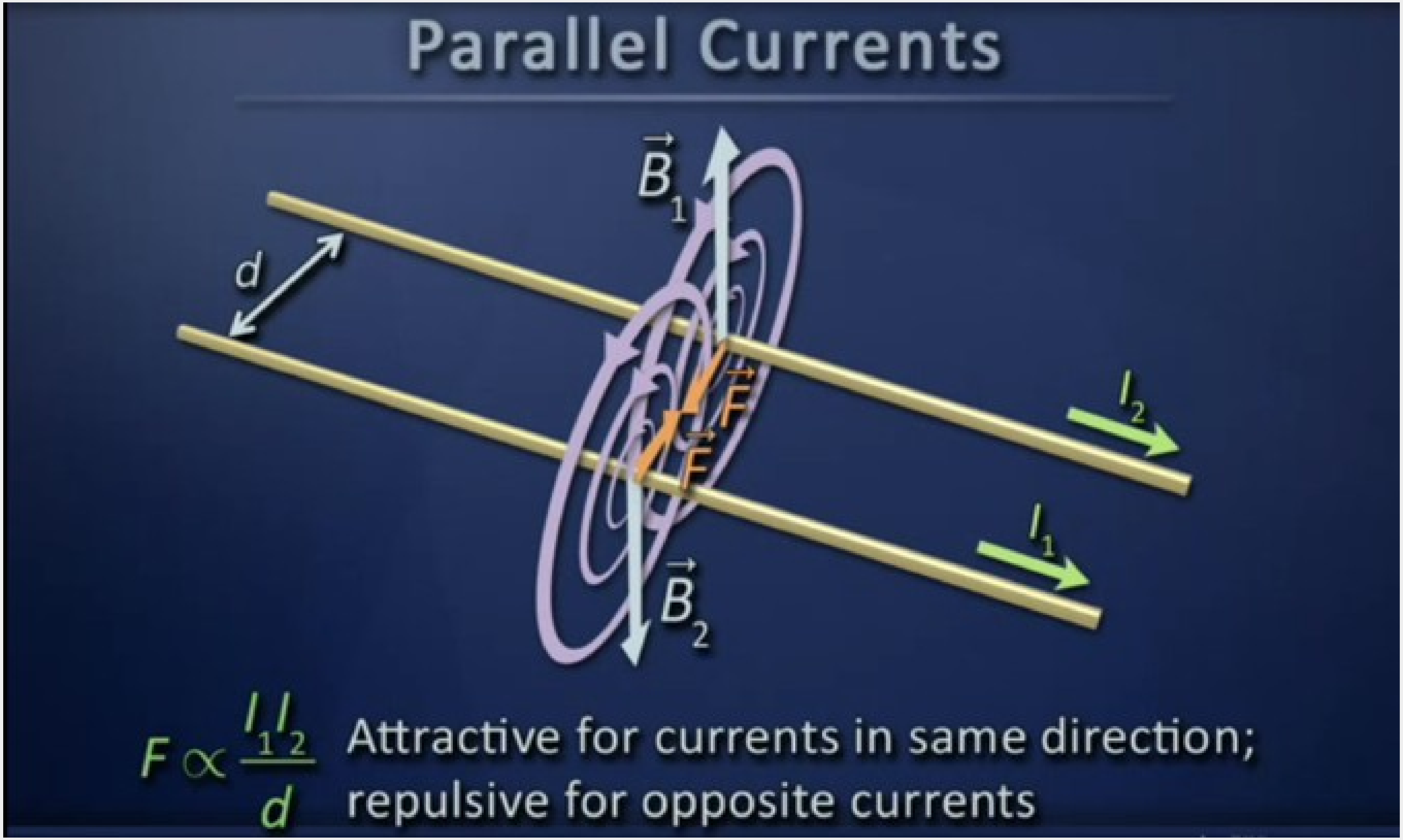
Reference:

https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_University_Physics_(OpenStax)/Book%3A_University_Physics_II_-_Thermodynamics_Electricity_and_Magnetism_(OpenStax)/12%3A_Sources_of_Magnetic_Fields/12.03%3A_Magnetic_Field_due_to_a_Thin_Straight_Wire

Force between 2 wires.

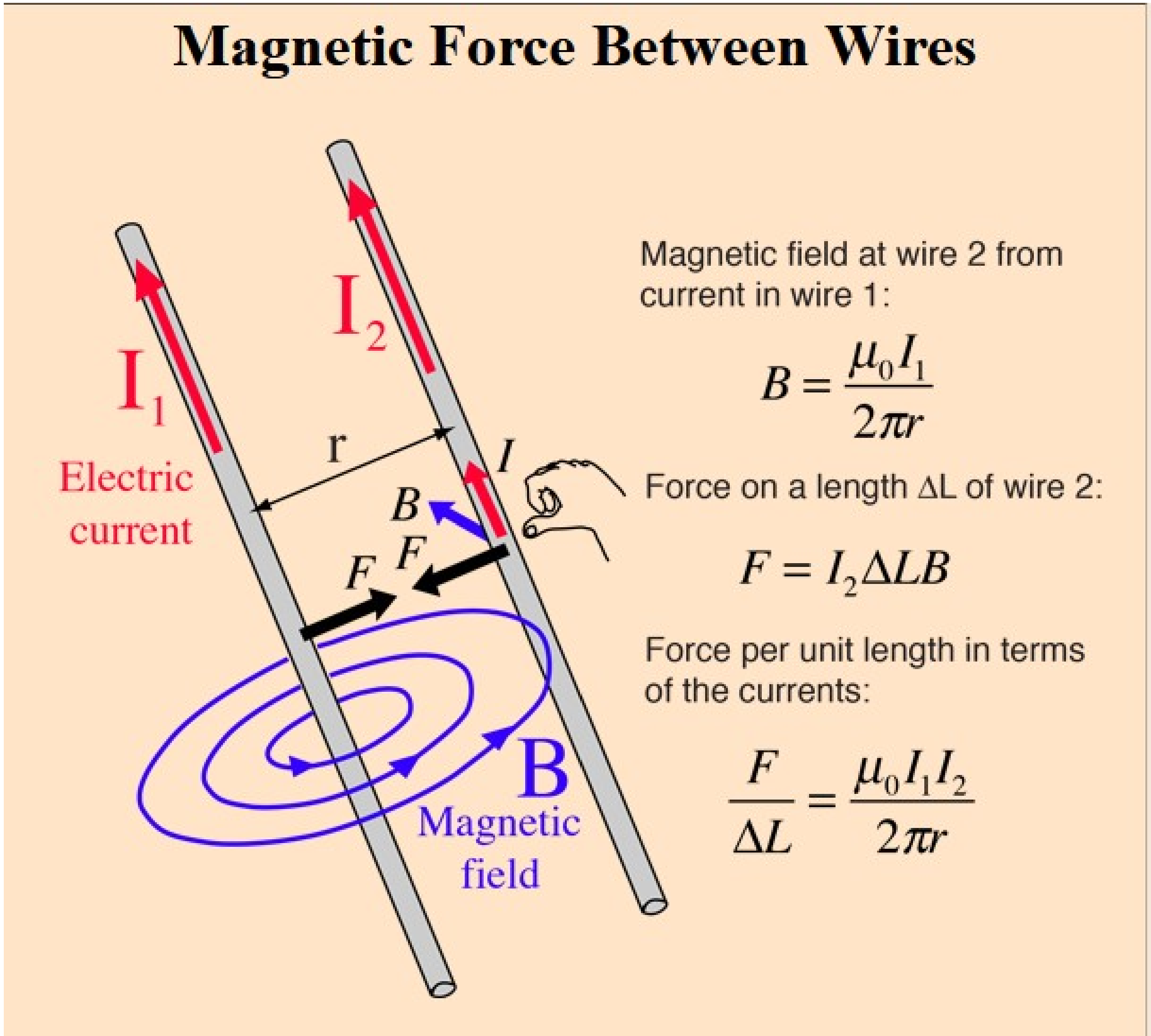


I1 → magnetic field and I2 responds to this field. The forces are equal in Magnitude (Newton's third law). So the force is proportional to I1I2 And inversely proportional to r. Video 2 wires.

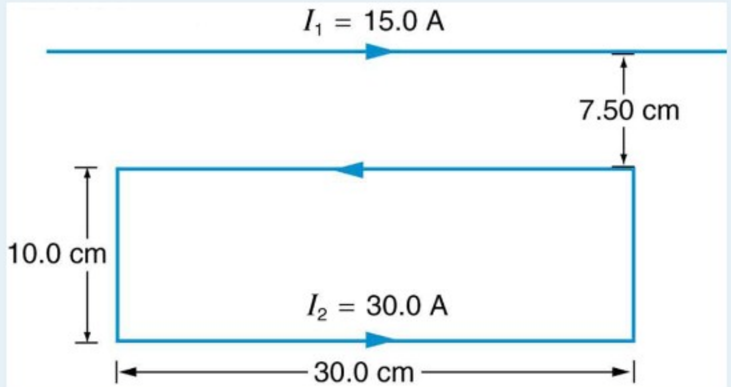


Show how to derive the expression. This explains the electric equipment humming. With AC the wires vibrate in transformer. Thats how 1A is defined

http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/wirfor.html http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/wirfor.html



6. Figure 4 shows a long straight wire near a rectangular current loop. What is the direction and magnitude of the total force on the loop?

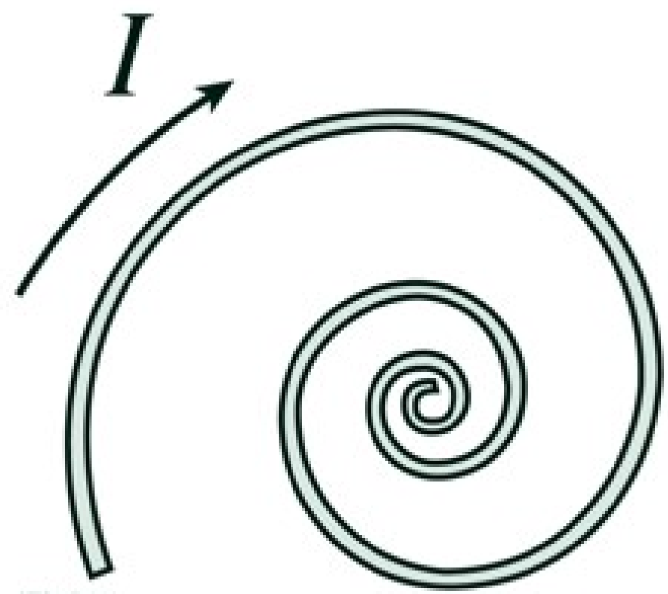


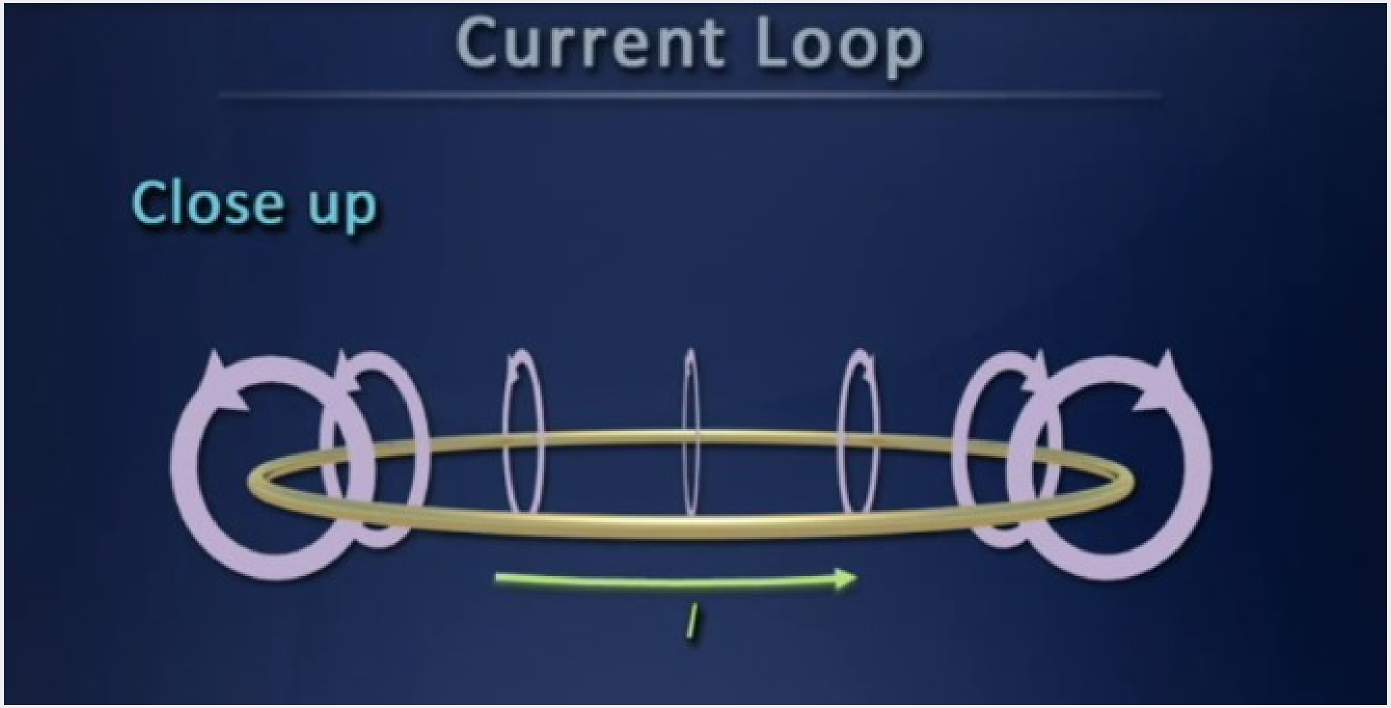
https://courses.lumenlearning.com/suny-physics/chapter/22-10-magnetic-force-between-two-parallel-conductors/

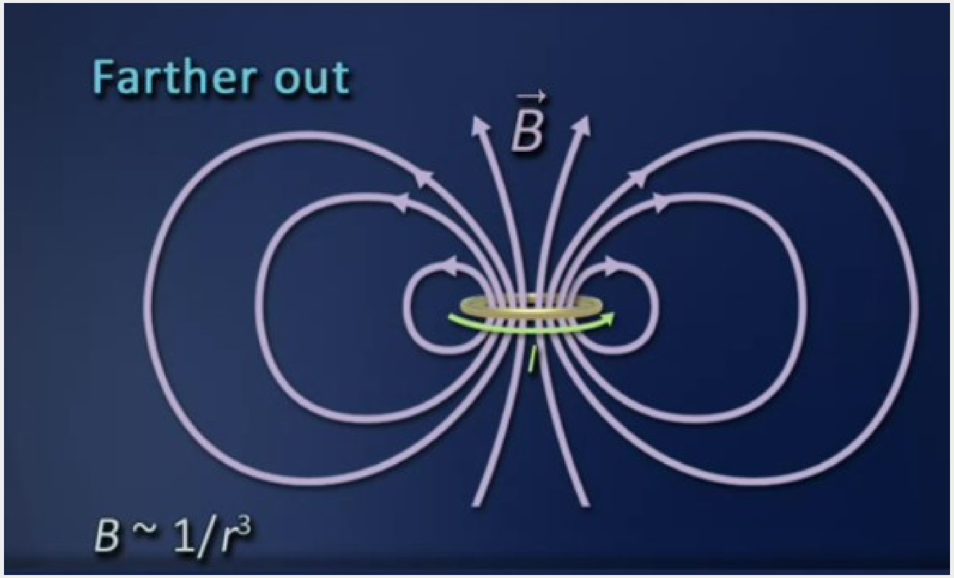
- A flexible wire is wound into a flat spiral as shown in the figure. If a current flows in the direction shown, will the coil tighten or loosen?

A. The coil will tighten.

B. The coil will loosen.







Video dipole → 1:40
Loop and toroid.

The magnetic field decreases
With $1/r^3$

Magnetic dipole

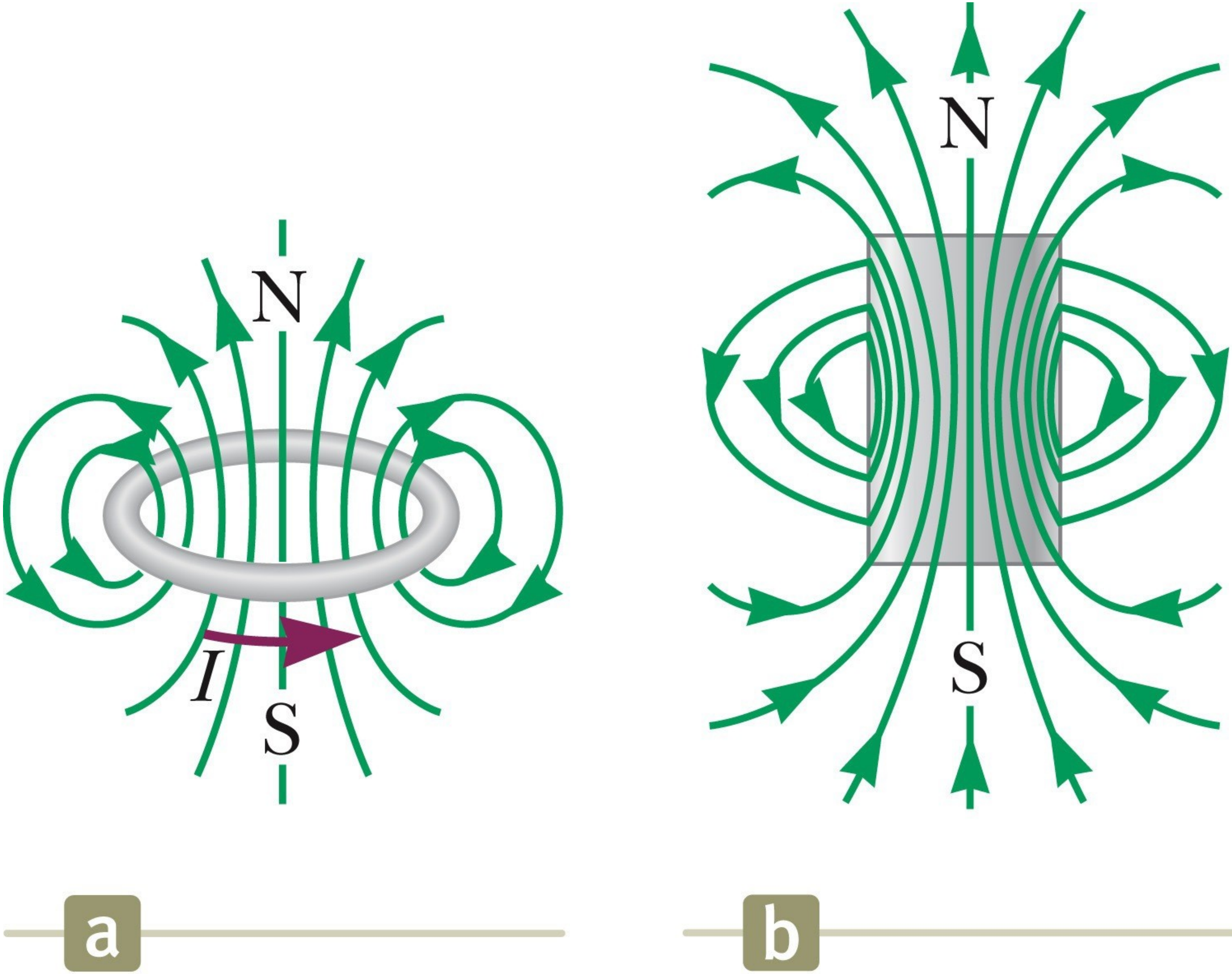
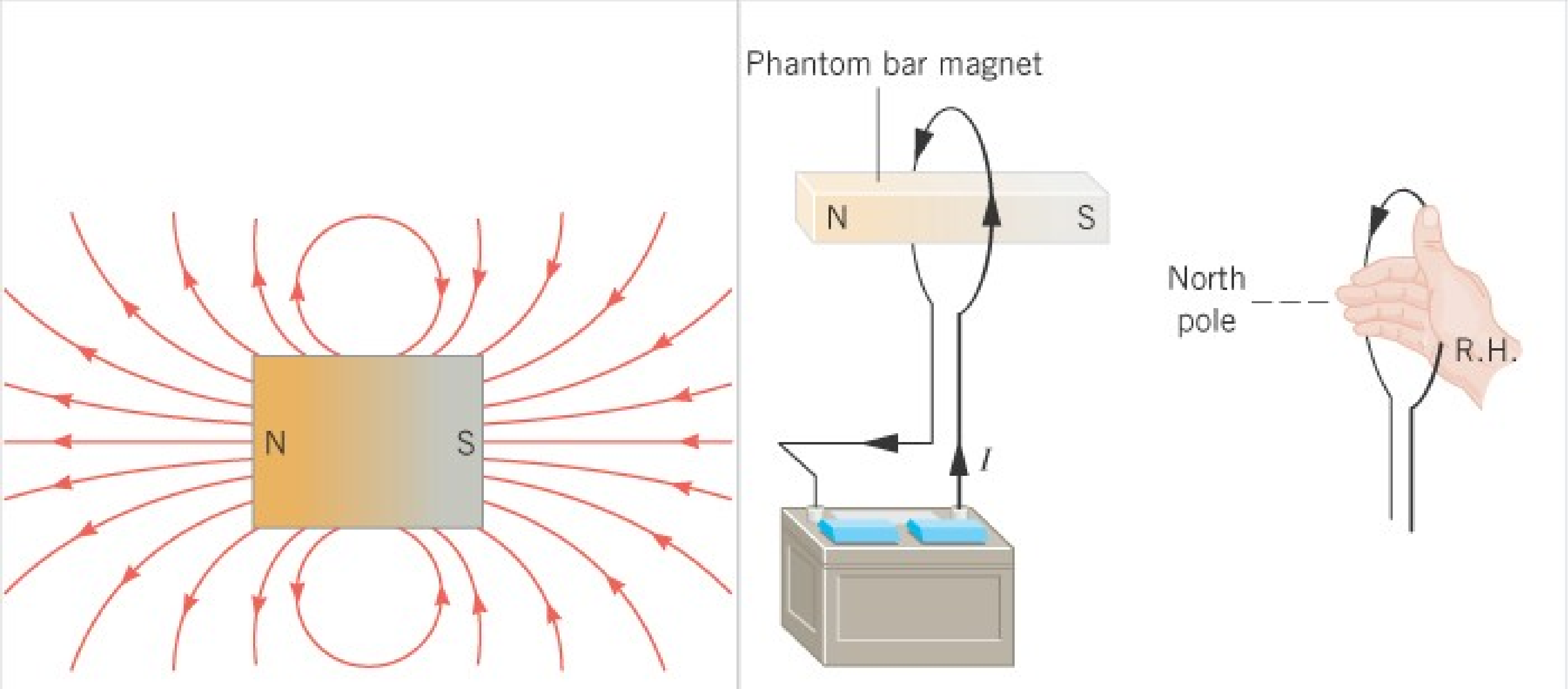
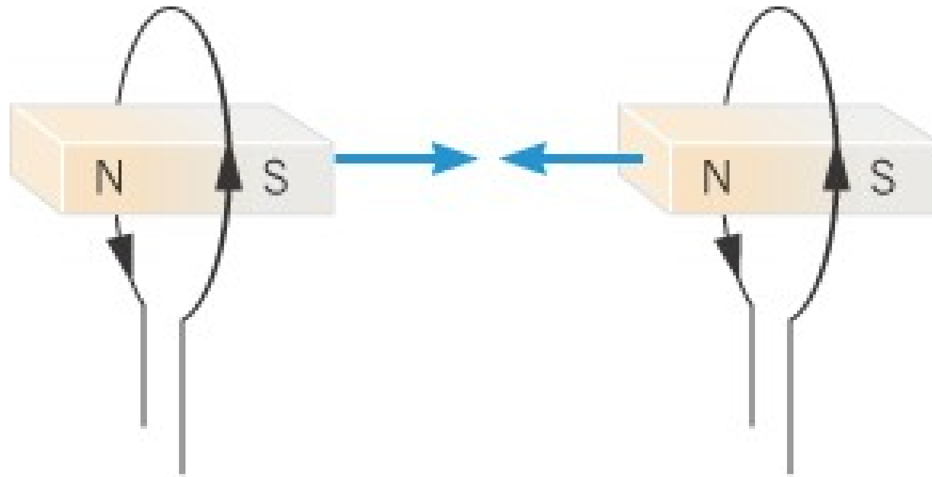


Figure 30-6 p909

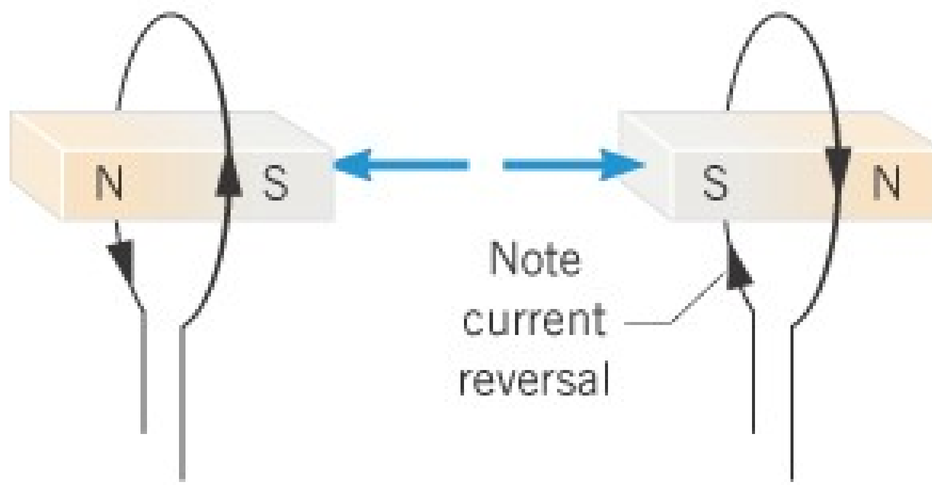
The field lines around the bar magnet resemble those around the loop.



(a)
(b)



(a) Attraction



(b) Repulsion

A LOOP OF WIRE

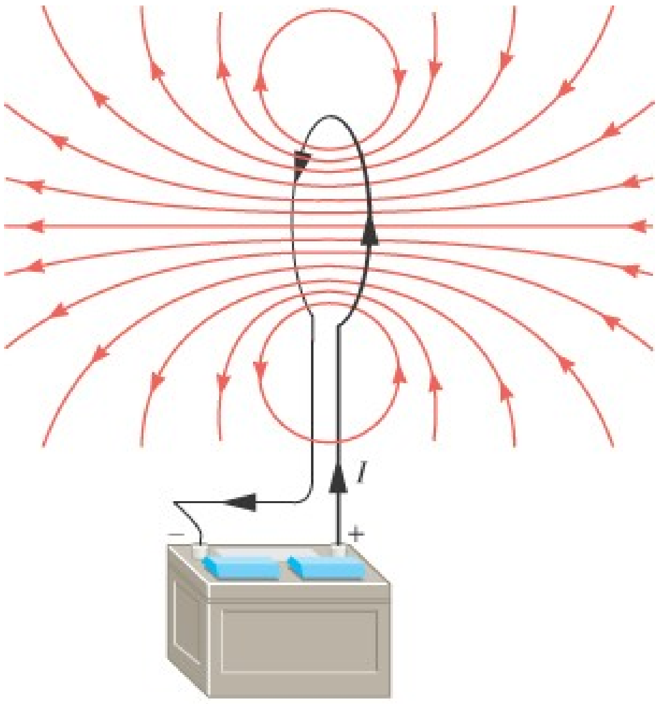
Magnetic field at the center of the loop

$$B = \frac{\mu_o I}{2R}$$

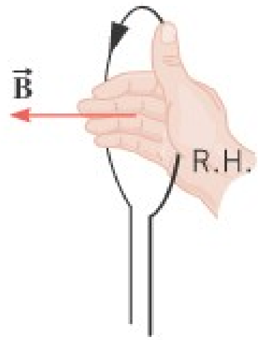
center of circular loop

Use Biot-Savart to show it.

http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/curloo.html



(a)



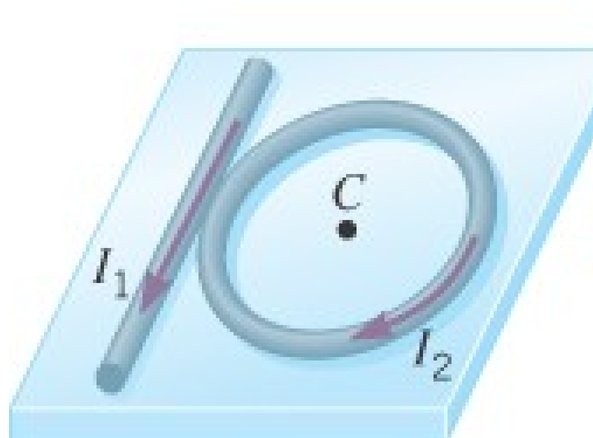
(b)

Example 10 Finding the Net Magnetic Field

A long straight wire carries a current of 8.0 A and a circular loop of wire carries a current of 2.0 A and has a radius of 0.030 m. Find the magnitude and direction of the magnetic field at the center of the loop.

$$B = \frac{\mu_o I}{2\pi r}$$ Long wire

$$B = \frac{\mu_o I}{2R}$$ loop

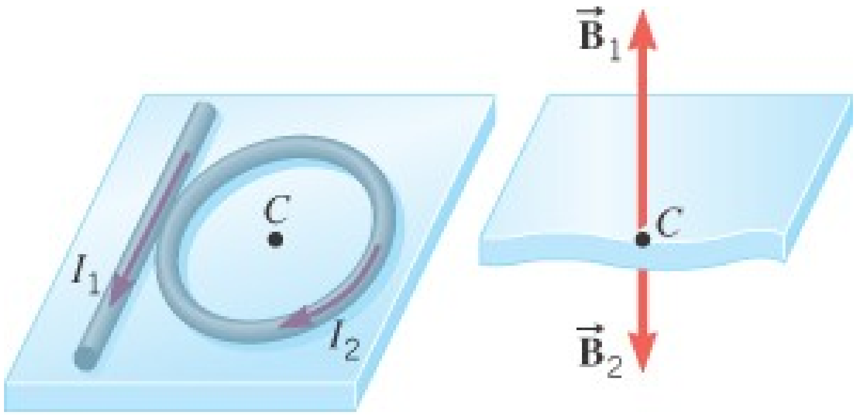


(a)

Find the net sum
Its a vector sum

(b)

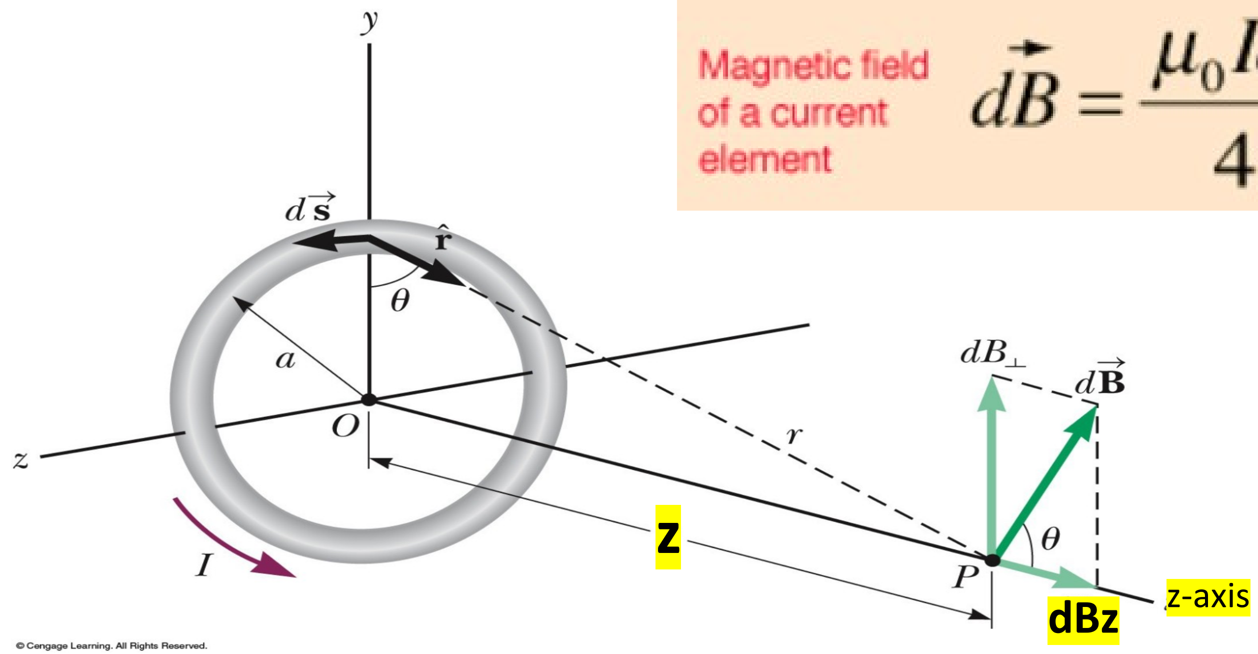
$$B = \frac{\mu_o I_1}{2\pi r} - \frac{\mu_o I_2}{2R} = \frac{\mu_o}{2}\left(\frac{I_1}{\pi r} - \frac{I_2}{R}\right)$$



(a) (b)

$$B = \frac{\left(4\pi\times10^{-7}\,T\cdot m/A\right)}{2}\left(\frac{8.0\ \mathrm{A}}{\pi\left(0.030\ \mathrm{m}\right)} - \frac{2.0\ \mathrm{A}}{0.030\ \mathrm{m}}\right) = 1.1\times10^{-5}\,T$$

Magnetic field at a distance z from the center → use biot- savart law and calculus
USE YOUR right hand. B is along the z axis.



Magnetic field of a current element

$$d\vec{B} = \frac{\mu_0 I d\vec{L} \times \vec{1}_r}{4\pi r^2}$$

$$B_Z = \frac{\mu_0}{4\pi} \frac{2\pi R^2 I}{\left(z^2 + R^2\right)^{3/2}}$$

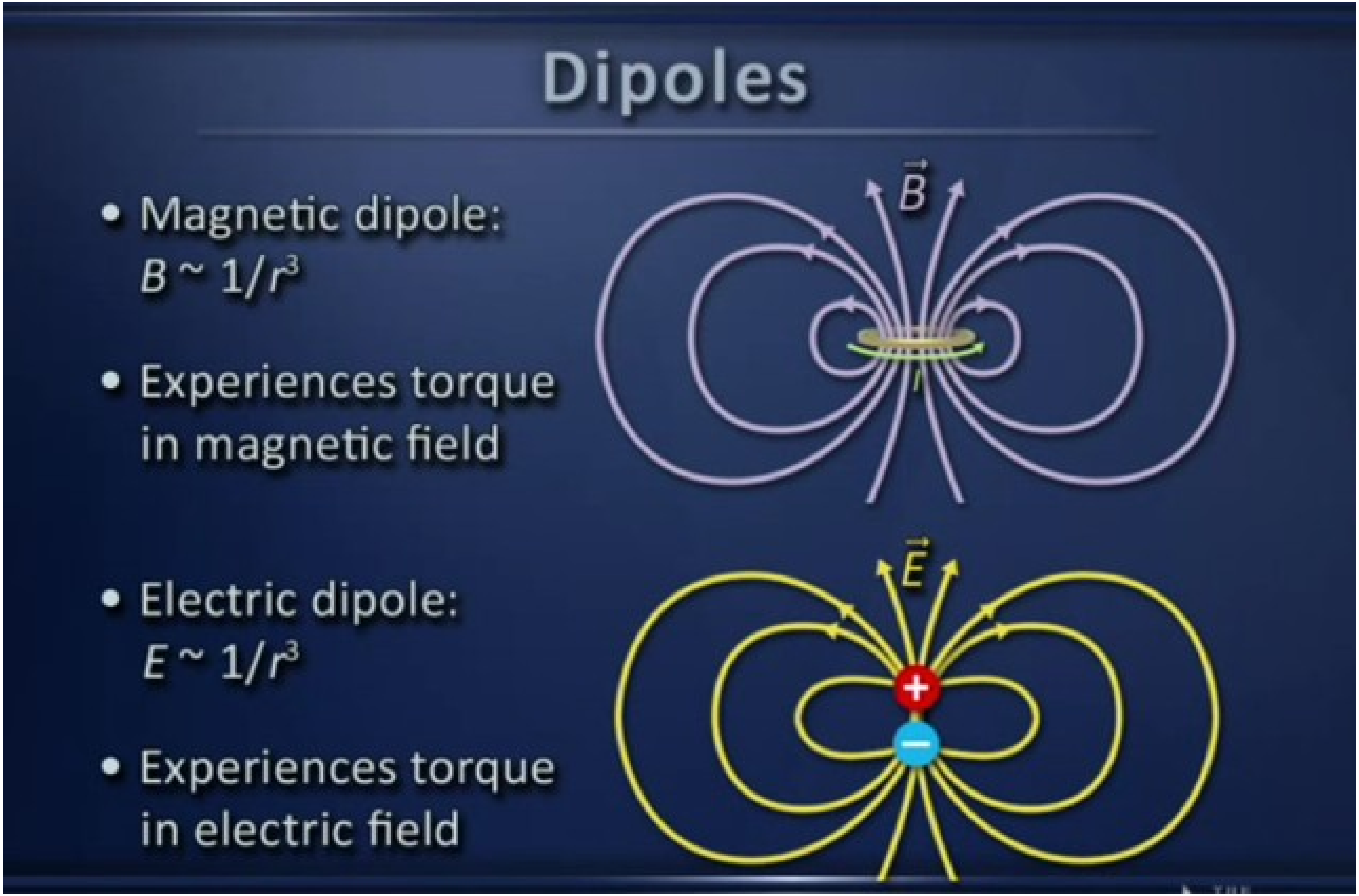
If z = 0 we find the B at the center of the loop.
B=uo I / 2R

What happens when z >>R ??
Very far away.

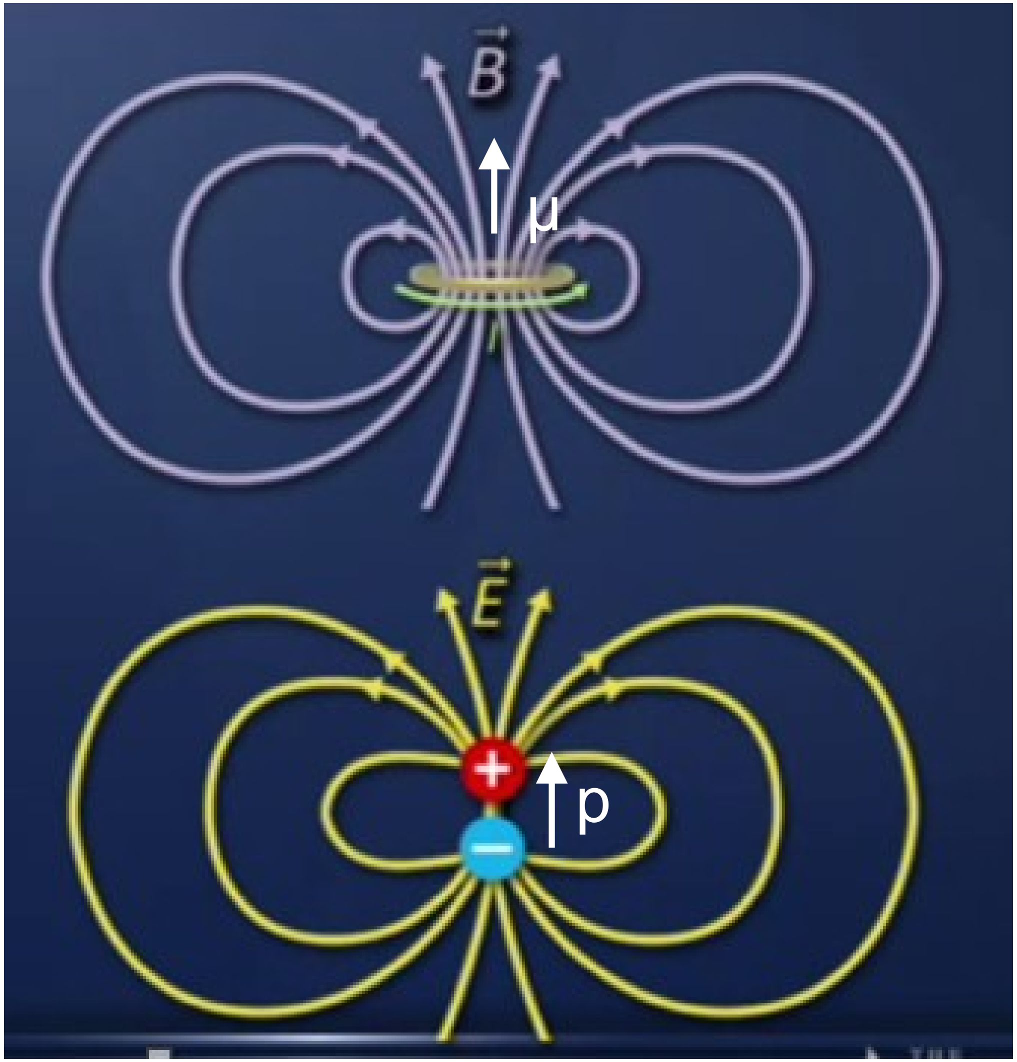
http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/curloo.html

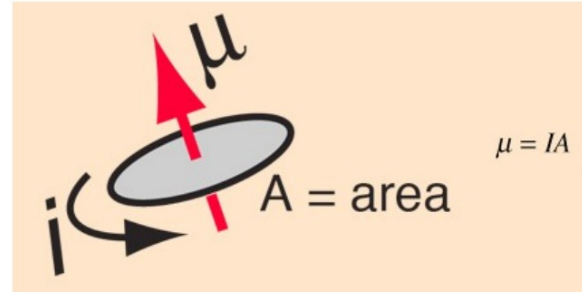
Video dipole

Electric dipole vs magnetic dipole



At a large distance >> size of the loop/dipole





$$, B = \mu_0 I (\pi R^2)/(2 \pi z^3).$$

$$= 2 \mu \mu_0/(4 \pi z^3), \text{ where } \mu = IA,$$

$$\vec{p} = q\vec{d}$$

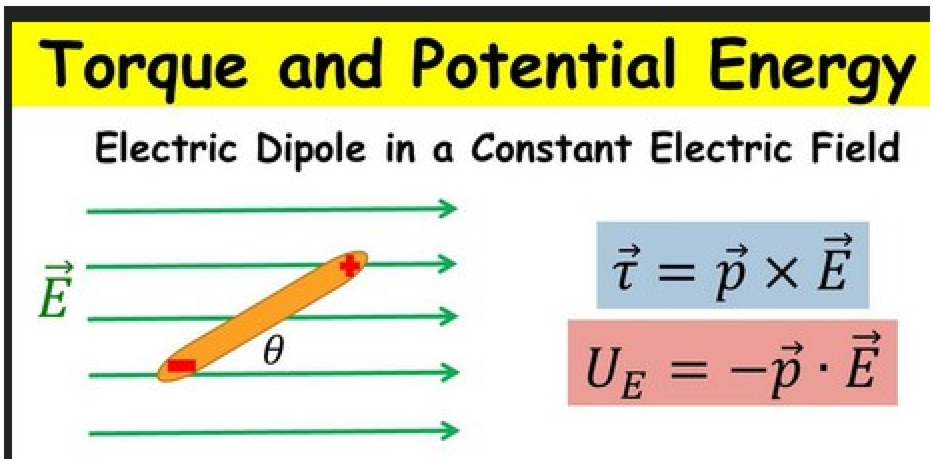
$$\vec{E}_{dipole} = \frac{1}{4\pi\varepsilon_0}\frac{-\vec{p}}{z^3}$$

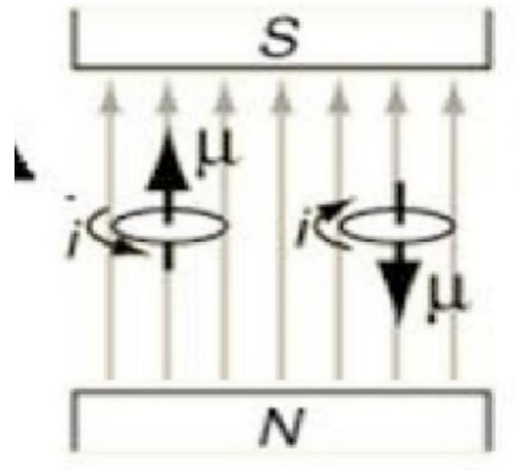
https://www.physics.unlv.edu/~lenz/PHYS%20181.S08/chap32sum.htm

See MRI application

https://phet.colorado.edu/en/simulations/mri

$$\tau = \mu \ \ x \ \ B$$

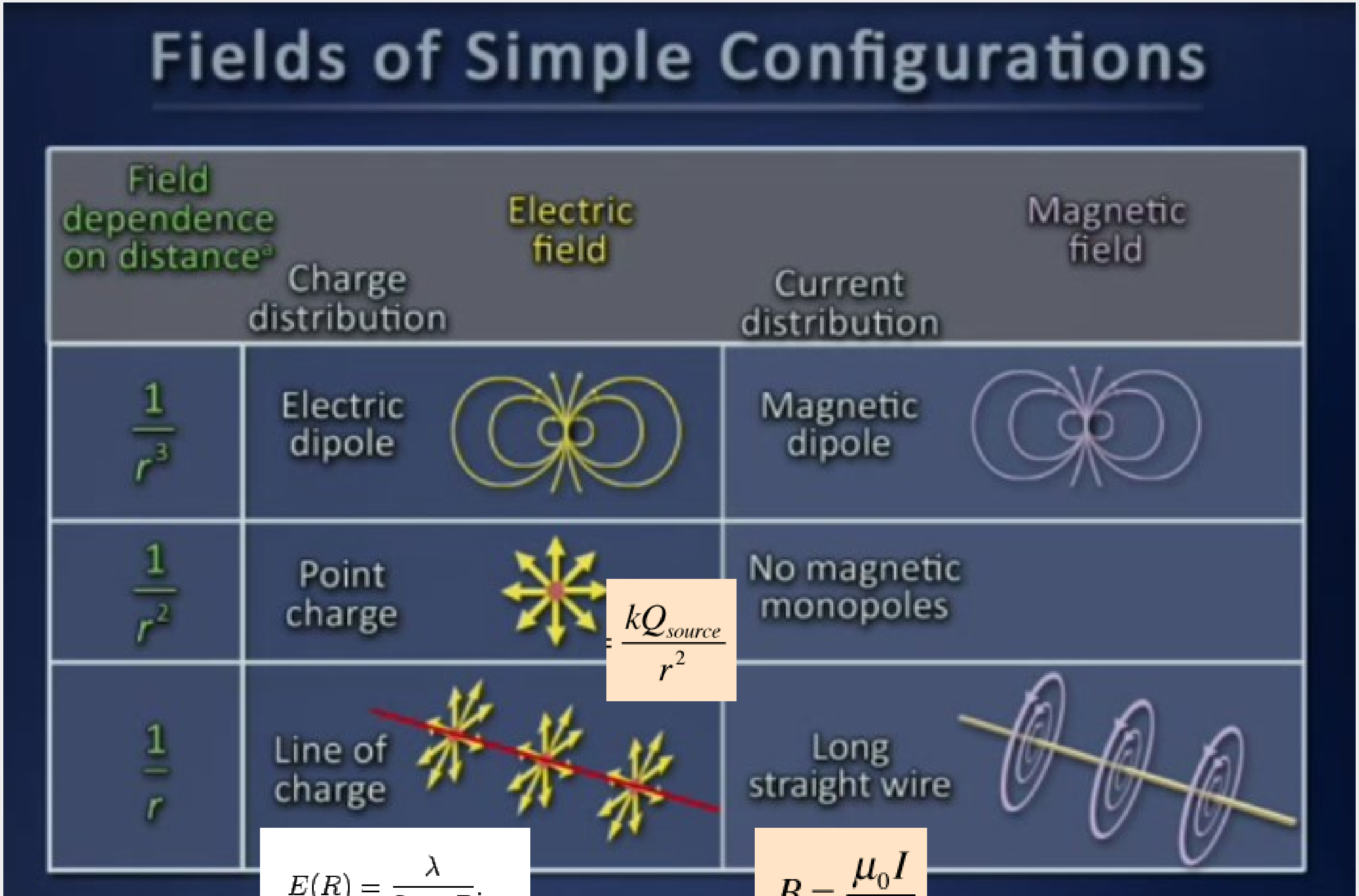




$$U(\theta) = -\mu \cdot B$$

An electric field produces a torque on an electric dipole **p**, **torque** = **p** x **E**.	A magnetic field produces a torque on a magnetic dipole, **μ**, **torque** = **μ** x **B**

http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/magmom.html



$$= \frac{kQ_{source}}{r^2}$$

$$E(R) = \frac{\lambda}{2\pi\epsilon_0 R}.$$

$$B = \frac{\mu_0 I}{2\pi r}$$

Simplest magnetic entity is a loop.
Simplest electric entity is a charge.
Both dipoles and loops respond to a field and experience a torque
Torque = pxE or torque =uxB See MRI application

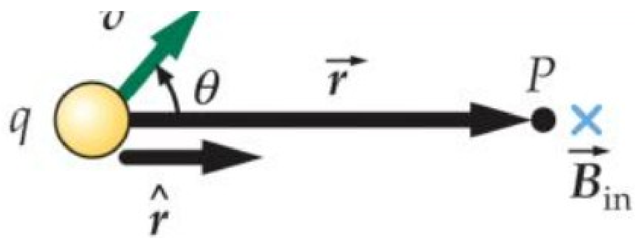
The Biot-Savart Law

A point charge produces an electric field. When the charge moves it produces a magnetic field, B:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{q\mathbf{v}\times\mathbf{r}}{|\mathbf{r}|^3}$$

μ_0 is the magnetic constant:

$$\boxed{\begin{aligned}\mu_0 &= 4\pi\times10^{-7}\,\mathrm{T\cdot m/A}\\ &= 4\pi\times10^{-7}\,\mathrm{N/A^2}\end{aligned}}$$



As drawn, the field is into the page

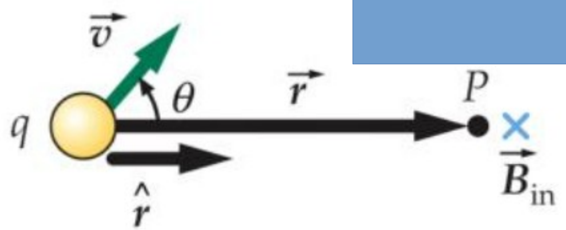
$$\boxed{\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}}$$

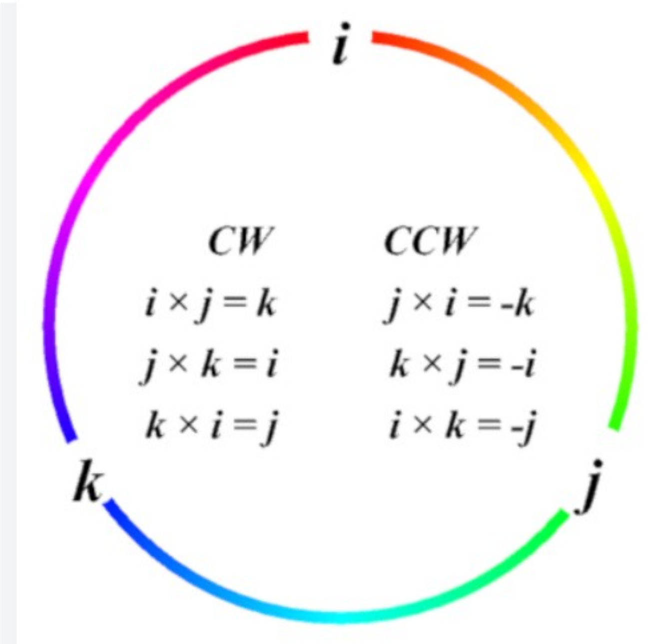
Example: $\vec{v} = 5\,\text{m/s}\,\hat{k}$

$\vec{r} = 3\,\text{m/}\,\hat{i} + 4\,\text{m/}\,\hat{j}$

$q = 4\,\mu\text{C}$

$\vec{B} =$?





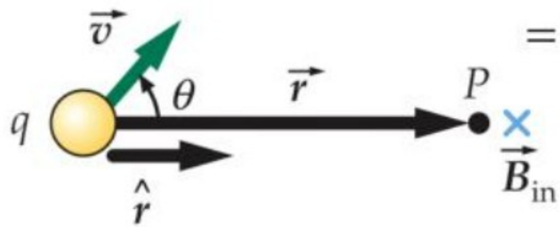
Example: $\vec{v} = 5\,\text{m/s}\,\hat{k}$

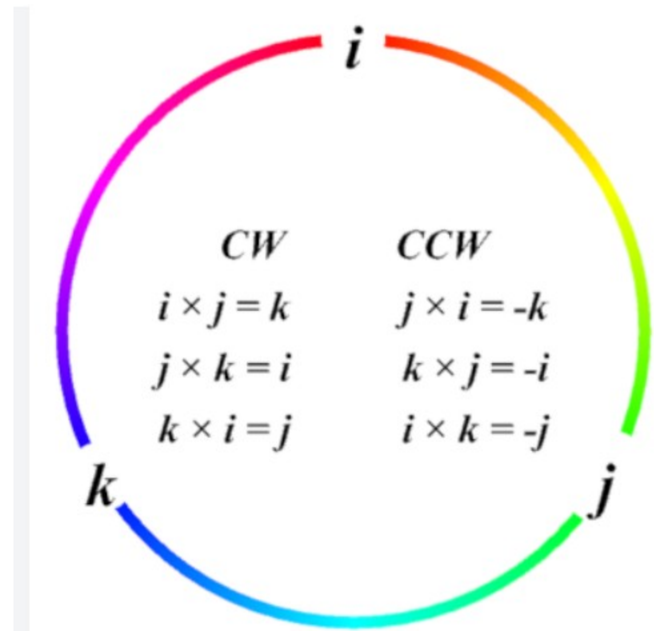
$$\vec{r} = 3\,\text{m/s}\,\hat{i} + 4\,\text{m/s}\,\hat{j}$$

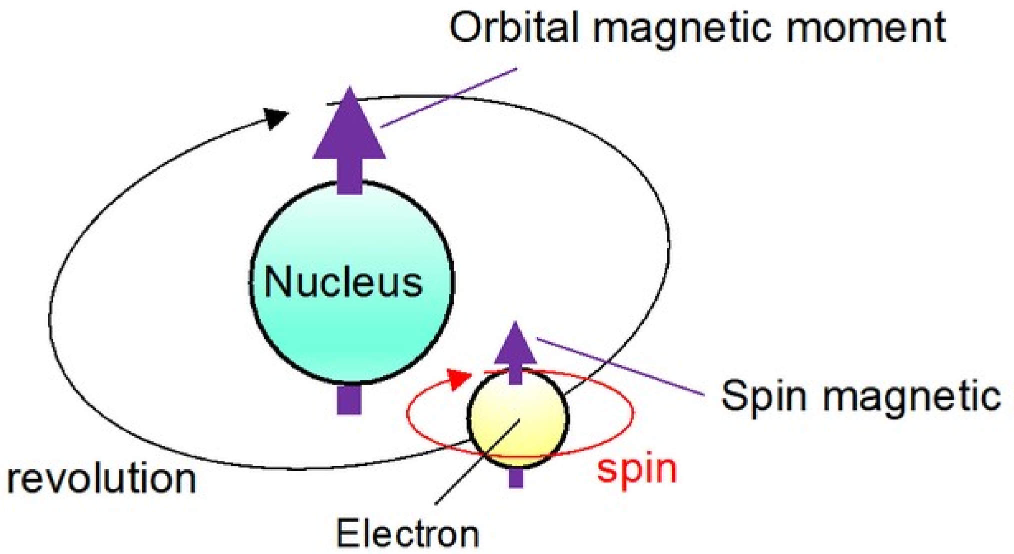
$$q = 4\,\mu\text{C}$$

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{(4\mu\text{C})(5\,\text{m/s}\,\hat{k})\times(3\,\text{m/s}\,\hat{i} + 4\,\text{m/s}\,\hat{j})}{(5\,\text{m/s})^3}$$

$$= (4.8\,\hat{j} - 6.4\,\hat{i})\times 10^{-14}\,\text{T}$$







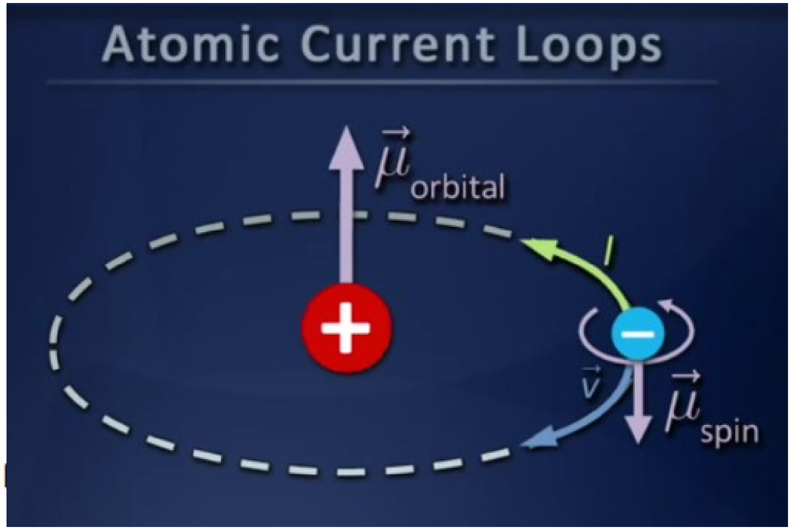


Fig. 1, source: https://group.ntt/en/newsrelease/2019/02/12/190212a.html

Origin of magnetism

The spin of the electrons in atoms is the main source of ferromagnetism, although **there is also a contribution from the orbital angular momentum of the electron about the nucleus**.

Spins of proton and electron are equal. But the magnetic moments are inversely proportional to the mass of the particle. **Proton is about 2000 times heavier than the electron** so proton's magnetic moment is smaller by three orders of magnitude. Thus its effect is negligible in comparison to electron magnetic moment.

Three Types of Magnetism

Ferromagnetism

- Strong interaction among atomic dipoles causes them to align, resulting in strong, large-scale magnetization
- Strong attraction to magnets

Paramagnetism

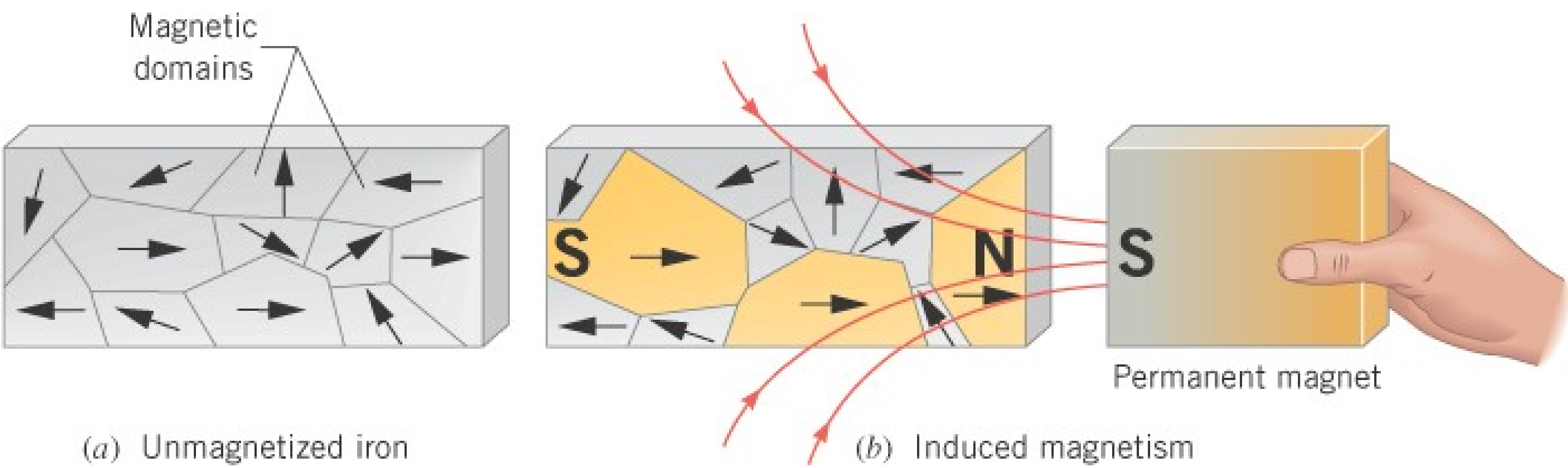
- Weak interaction among atomic dipoles
- Weak attraction to magnets

Diamagnetism

- Induced atomic magnetic moments result in repulsion from magnets

The intrinsic “spin” and orbital motion of electrons gives rise to the magnetic properties of materials.

In ***ferromagnetic materials*** groups of neighboring atoms, forming ***magnetic domains,*** the spins of electrons are naturally aligned with each other.



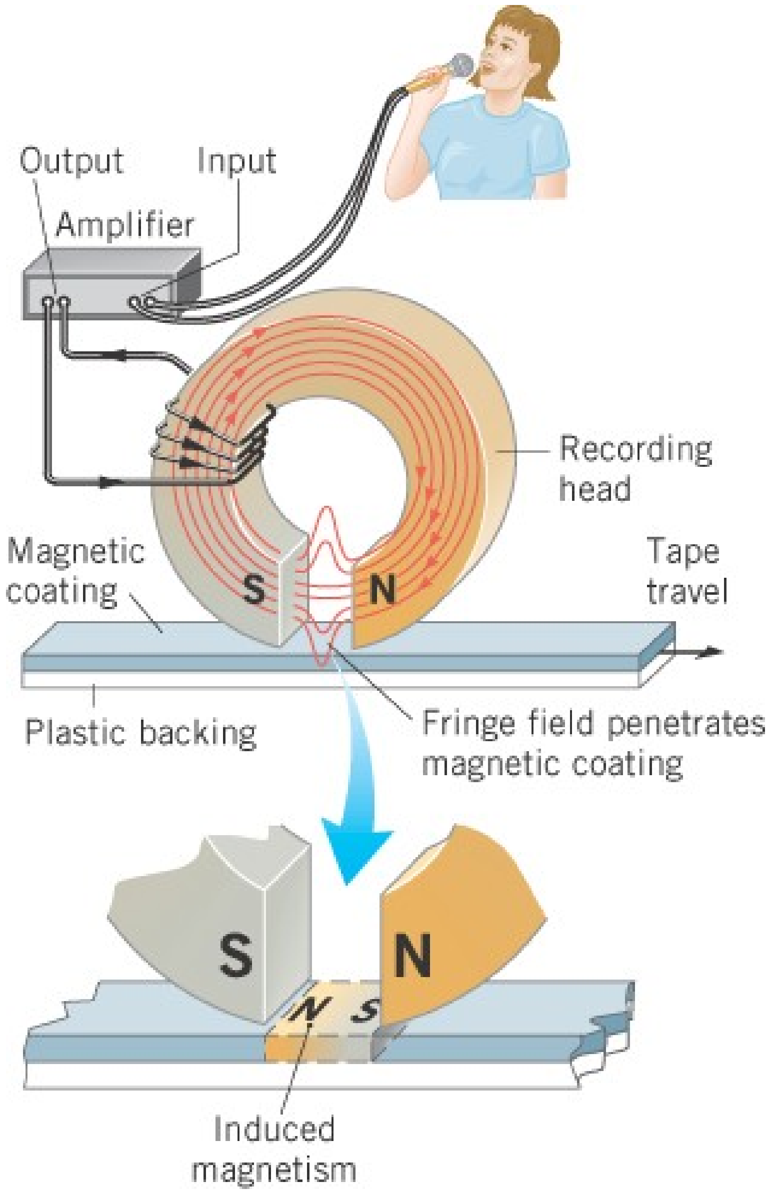
(*a*) Unmagnetized iron

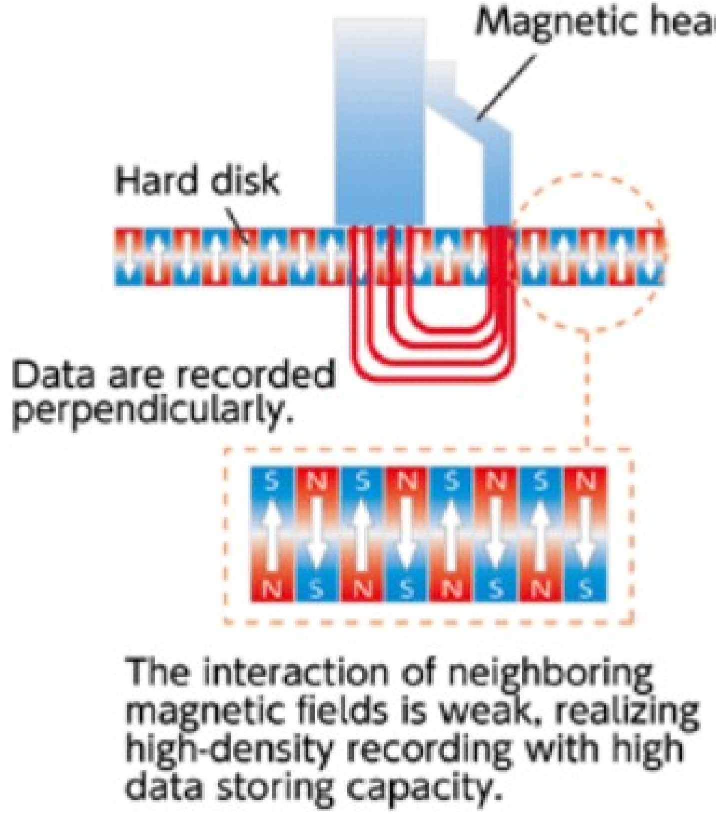
(*b*) Induced magnetism

You can hear
The domains
flipping

Curie temperature

21.9 Magnetic Materials

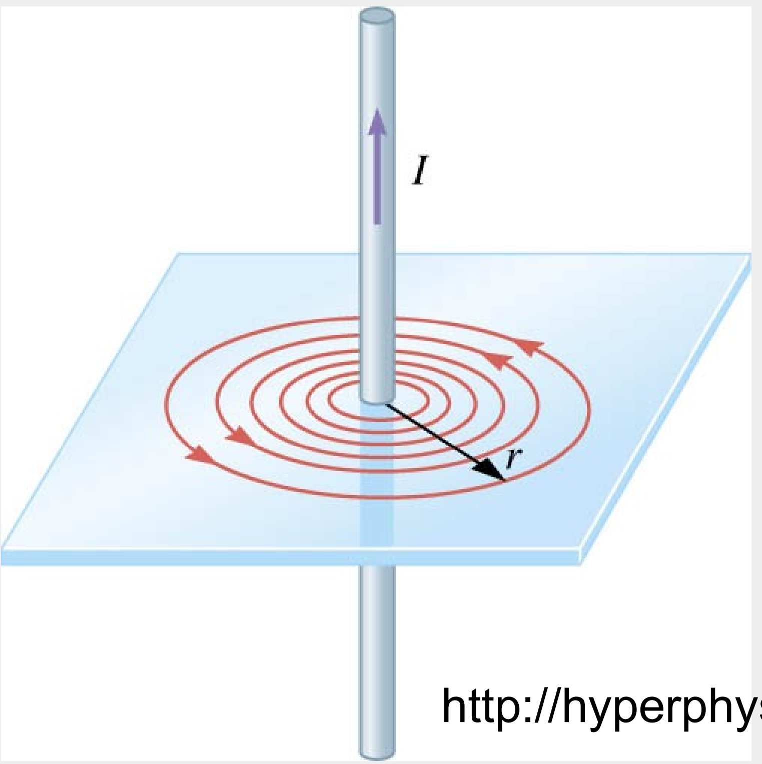


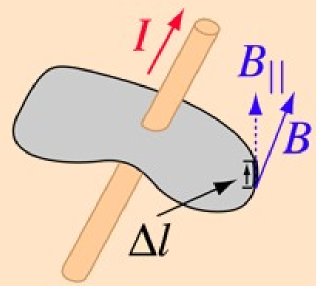


Video. B circulates.

Ampere’s law

$$B = \frac{\mu_o I}{2\pi r}$$





$$\sum B_{||} \Delta l = \mu_0 I$$

Multiply both side by 2 pi r (circumference)

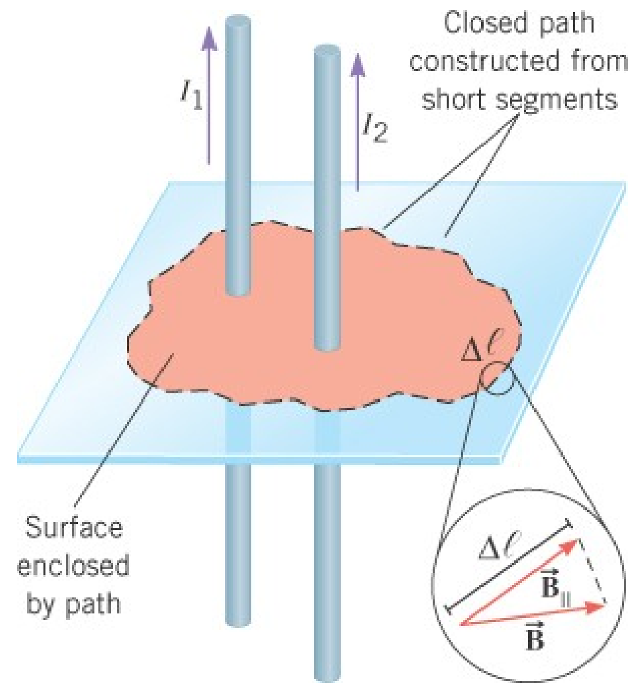
→ Amperes law.

→ B circulates on a closed path around
An open surface

http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/amplaw.html

AMPERE’S LAW FOR STATIC MAGNETIC FIELDS

For any current geometry that produces a magnetic field that does not change in time,

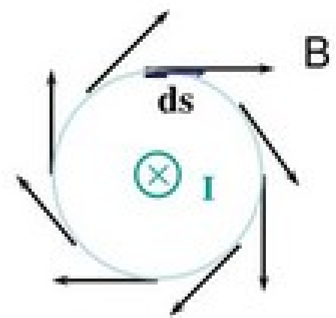


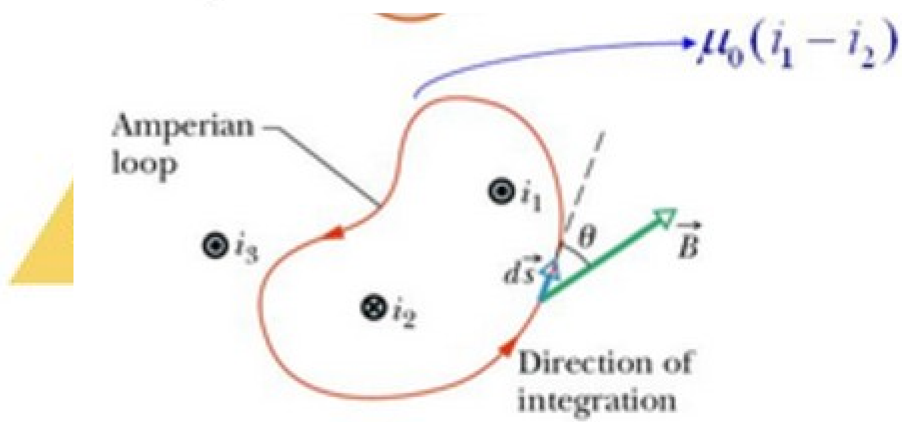
Ampere’s Law

- For small length elements *ds* on a closed path (not necessarily circular)

$$\oint \mathbf{B}\cdot d\mathbf{s} = \mu_o I$$

- *I* is enclosed current passing through any surface bounded by the closed path.
- ***Note: dot product***
- ***Use where there is high symmetry***





André-Marie Ampère was born on January 20, 1775 at Poleymieux, sixteen miles away from the city of Lyons, France to a wealthy business family during the height of French Enlightenment. His father, Jean-Jacques Ampère, was a great admirer of the philosophy of Jean-Jacques Rousseau who argued that education should be direct from nature. Since Jean-Jacques Ampère did not believe in formal schooling, he actualized this idea by educating Ampere within the walls of his well-stocked library.

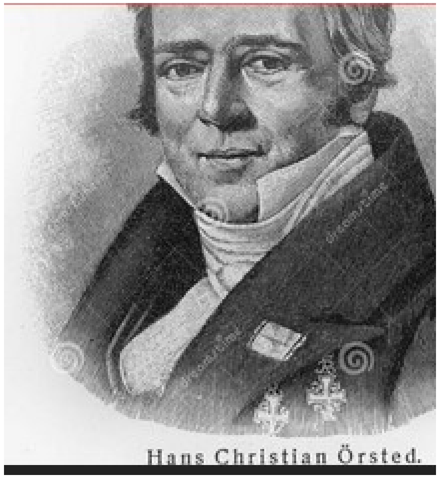
Interesting story. His dad was guillotined and left him a library.
He taught himself Math and Sciences. His first wife dies. No luck with his kids.
Divorced his second wife. Almost discovered Iodine, Chlorine, Avogadro law.
From Chemist → Mathematician. Got interested in Oersted experiment.
(Science can happen outside France).

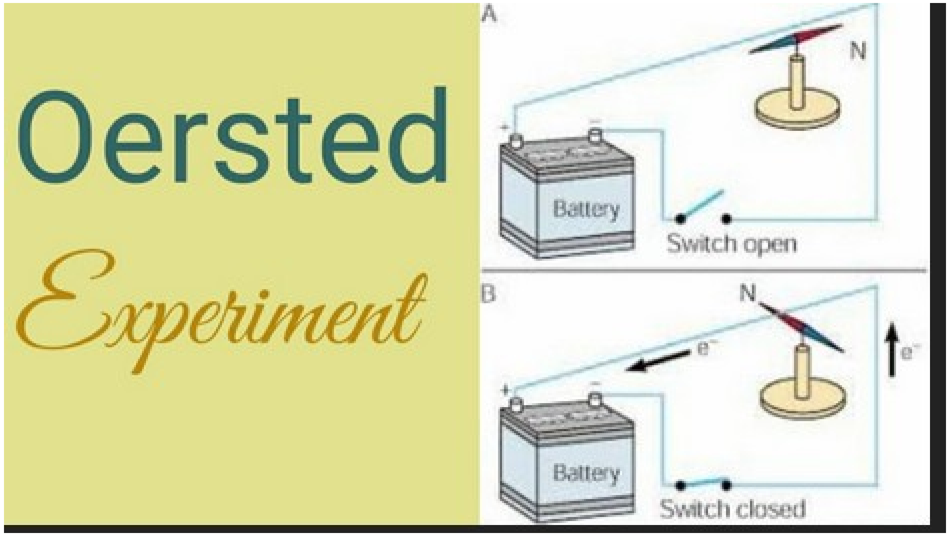
1820→ His law. Father of electrodynamics.

https://www.upsbatterycenter.com/blog/andre-marie-ampere/

He was usually considered as the first person to discover electromagnetism. He is credited for the invention of the astatic needle which is the vital component of the static galvanometer. He was also the first one to determine that when two wires are charged with electricity, it generates a magnetic field.

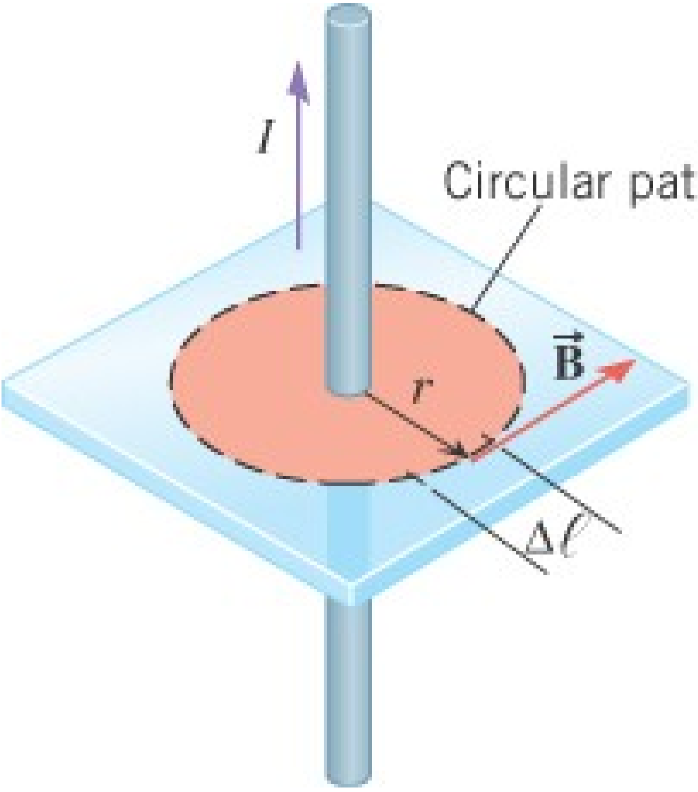
In 1820, H.C Ørsted accidentally discovered the phenomenon establishing the relationship between magnetism and electricity. Ampère performed a series of experiments to clarify the exact nature of its relationship and performed his famous formula law on electromagnetism which was later known as the Ampère law. And his findings were reported in the Académie des Sciences one week after Ørsted's discovery. His theory in electromagnetism was his greatest contribution.





Example 11 An Infinitely Long, Straight, Current-Carrying Wire

Use Ampere's law to obtain the magnetic field.



$$\sum B_{||}\Delta\ell = \mu_o I$$

$$\Downarrow$$

$$B\left(\sum \Delta\ell\right) = \mu_o I$$

$$\Downarrow$$

$$B 2\pi r = \mu_o I$$

$$\Downarrow$$

$$B = \frac{\mu_o I}{2\pi r}$$

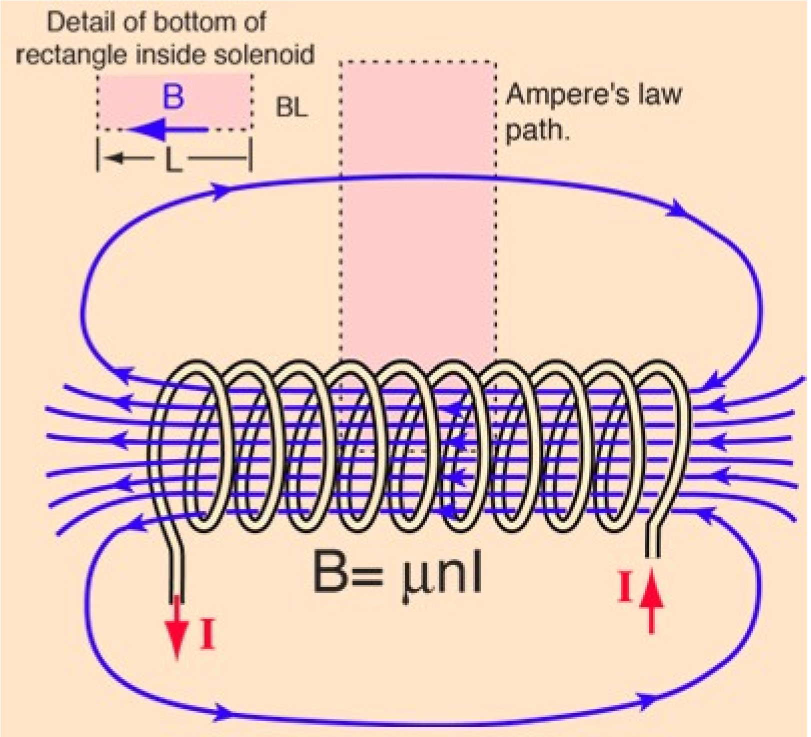
A SOLENOID

http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/solenoid.html#c2

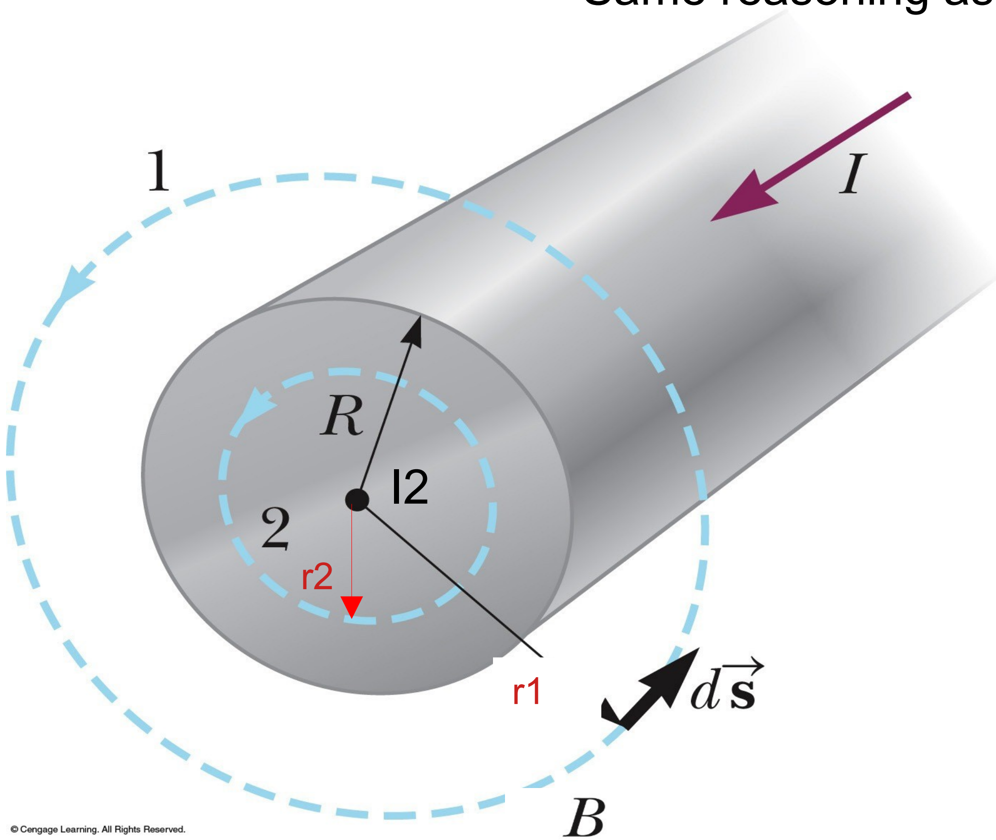
number of turns per unit length

Interior of a solenoid

$$B = \mu_o n I$$



Same reasoning as in mechanical Eng: (take water)



Area of the cross-section of cable Radius R is A.
Area of the cross-section of imaginary cylinder radius r2 is A2

Logic: I2 / I = A2/A EQ1

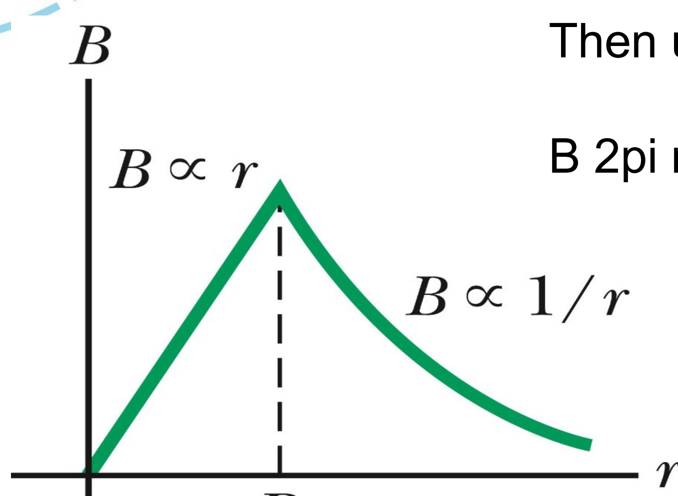
Then use amperes law
r<R
B 2pi r2 = uo (I2) EQ2

$$B = \frac{\mu_0 I r}{2\pi R^2}$$

Then use amperes law r>R

B 2pi r1 = uo I

$$B = \frac{\mu_0 I}{2\pi r}$$

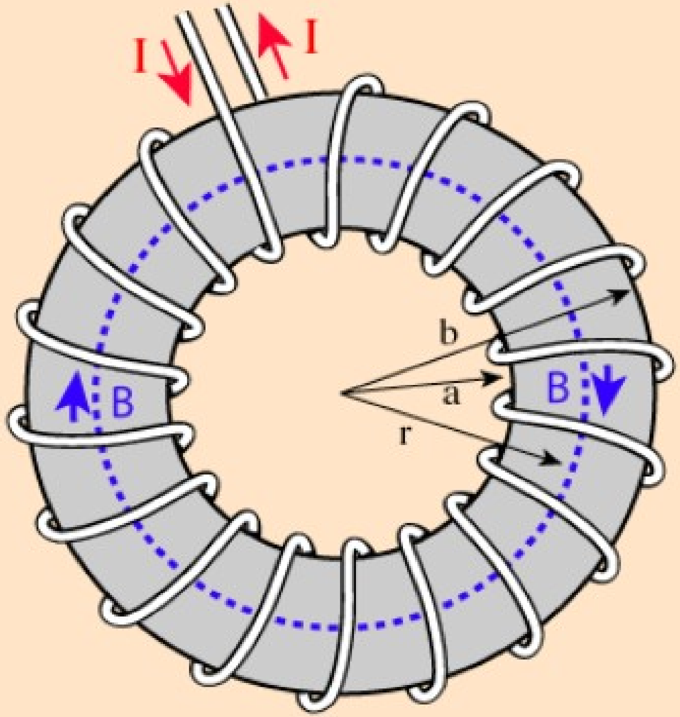


Show in class

http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/magin.html#c1

Figure 30-13 p913

Magnetic Field of Toroid



Finding the magnetic field inside a toroid is a good example of the power of Ampere's law. The current enclosed by the dashed line is just the number of loops times the current in each loop. Amperes law then gives the magnetic field by

$$B2\pi r = \mu NI$$

$$B = \frac{\mu NI}{2\pi r}$$

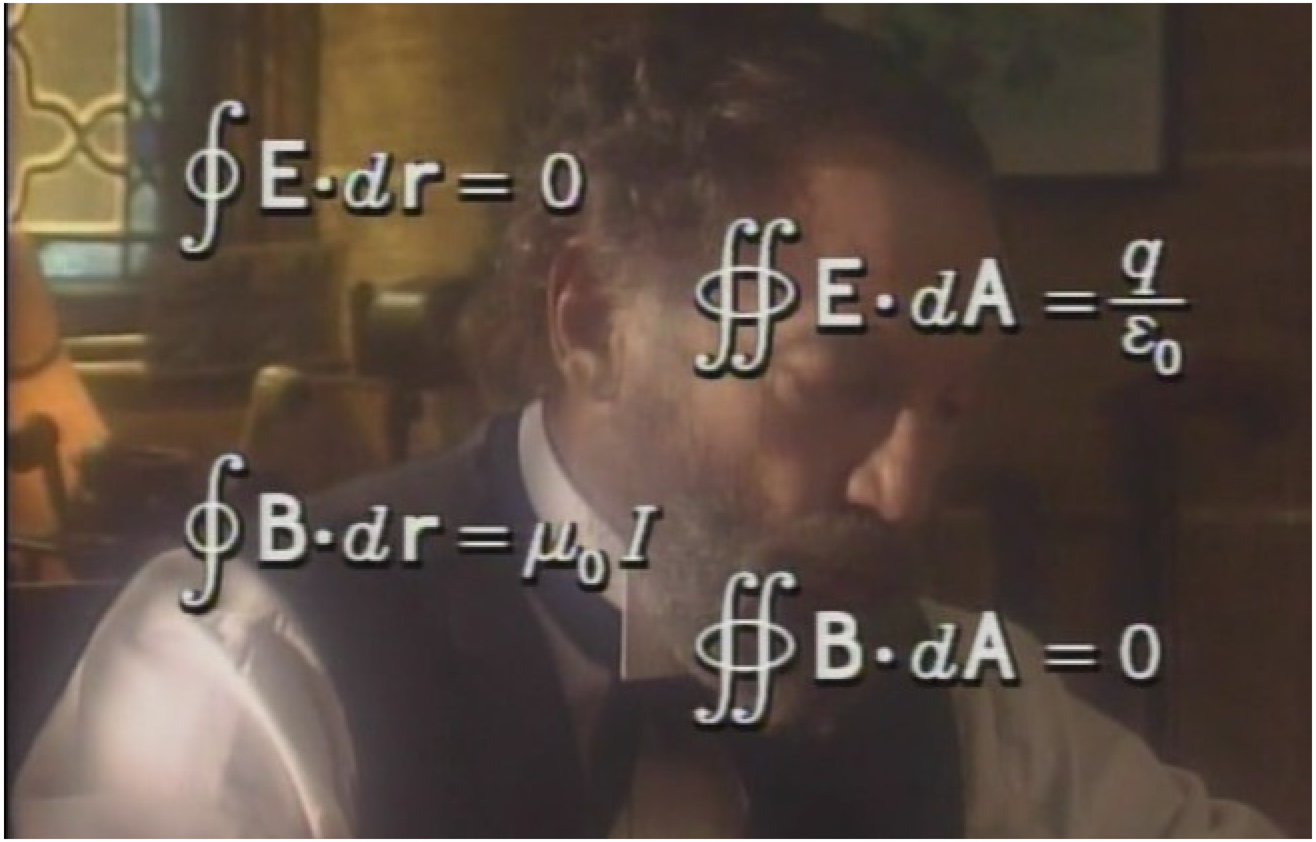
The toroid is a useful device used in everything from tape heads to tokamaks.

http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/toroid.html

Figure 30-15 p914

Electric Field	Magnetic Field
Charges, moving or stationary, create electric fields.	Moving charges (currents) create magnetic fields.
Elementary unit of charge is the point charge.	Elementary unit of magnetism is the dipole with a north and south pole. No one has ever seen an isolated north or south magnetic pole.
Electric field produces a force on a point charge, **F** = q **E**.	Magnetic field produces a force on a moving charge, **F** = q **vxB**. There is no magnetic force on stationary charges. Also the above equation is intrinsically three dimensional in that the force vector is perpendicular to the plane formed by **v** and **B**.
Coulomb's Law: **E** = q **r**/(4 $\pi\varepsilon_0$ r^2) where **r** is a unit vector pointing from the charge q producing the electric field to the point at which the electric field is being found.	Biot-Savart Law: **B** = μ_0 q **v** x **r**/(4 π r^2) where **v** is the velocity of the point charge producing the magnetic field. **r** is defined here as it was for Coulomb's Law.
Gauss' Law: ∫**EAn** ds = q/ε_0 Gauss's Law applies to a closed surface and q is the total charge inside the surface. The surface integral is the dot product of the electric field with the outward pointing normal vector, **n**, summed over the closed surface.	Ampere's Law: $\oint B \cdot ds = \mu_0 i$ Ampere's Law applies to a loop enclosing the current I. The path integral is the magnetic field dotted into the normalized tangent vector, **t**, to the curve forming the loop summed over the closed loop.
An electric field produces a torque on an electric dipole **p**, **torque** = **p** x **E**.	A magnetic field produces a torque on a magnetic dipole, μ, **torque** = μ x **B**

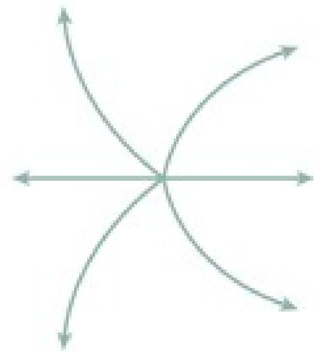
https://www.physics.unlv.edu/~lenz/PHYS%20181.S08/chap32sum.htm



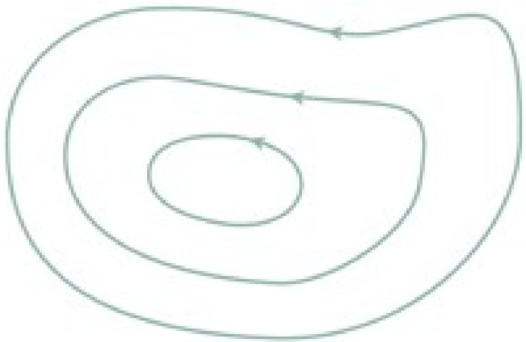
Maxwell equations for constant electric and magnetic fields.
The electric field diverge , the magnetic field circulates.
→ magnetic flux through a closed surface is 0 explain

Video Amperes law

- The figure shows two sets of field lines. Which set could be a magnetic field?
  A. (a)
  B. (b)



(a)



(b)

21.7 Magnetic Fields Produced by Currents An electric current produces a magnetic field, with different current geometries giving rise to different field patterns. For an infinitely long, straight wire, the magnetic field lines are circles centered on the wire, and their direction is given by Right-Hand Rule No. 2 (see below). The magnitude of the magnetic field at a radial distance r from the wire is given by Equation 21.5, where I is the current in the wire and μ_0 is a constant known as the permeability of free space ($\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T \cdot m/A}$).

$$B = \frac{\mu_0 I}{2\pi r} \tag{21.5}$$

Right-Hand Rule No. 2: Curl the fingers of the right hand into the shape of a half-circle. Point the thumb in the direction of the conventional current I, and the tips of the fingers will point in the direction of the magnetic field $\vec{\mathbf{B}}$.

The magnitude of the magnetic field at the center of a flat circular loop consisting of N turns, each of radius R and carrying a current I, is given by Equation 21.6. The loop has associated with it a north pole on one side and a south pole on the other side. The side of the loop that behaves like a north pole can be predicted by using Right-Hand Rule No. 2.

$$B = N\frac{\mu_0 I}{2R} \tag{21.6}$$

A solenoid is a coil of wire wound in the shape of a helix. Inside a long solenoid the magnetic field is nearly constant and has a magnitude that is given by Equation 21.7, where n is the number of turns per unit length of the solenoid and I is the current in the wire. One end of the solenoid behaves like a north pole, and the other end like a south pole. The end that is the north pole can be predicted by using Right-Hand Rule No. 2.

$$B = \mu_0 n I \tag{21.7}$$

21.8 Ampère's Law Ampère's law specifies the relationship between a current and its associated magnetic field. For any current geometry that produces a magnetic field that does not change in time, Ampère's law is given by Equation 21.8, where $\Delta\ell$ is a small segment of length along a closed path of arbitrary shape around the current, $B_\parallel$ is the component of the magnetic field parallel to $\Delta\ell$, I is the net current passing through the surface bounded by the path, and μ_0 is the permeability of free space. The symbol Σ indicates that the sum of all $B_\parallel\,\Delta\ell$ terms must be taken around the closed path.

$$\Sigma B_\parallel\,\Delta\ell = \mu_0 I \tag{21.8}$$

21.9 Magnetic Materials Ferromagnetic materials, such as iron, are made up of tiny regions called magnetic domains, each of which behaves as a small magnet. In an unmagnetized ferromagnetic material, the domains are randomly aligned. In a permanent magnet, many of the domains are aligned, and a high degree of magnetism results. An unmagnetized ferromagnetic material can be induced into becoming magnetized by placing it in an external magnetic field.